

GENERAL SIR JOHN KOTELAWALA DEFENCE UNIVERSITY

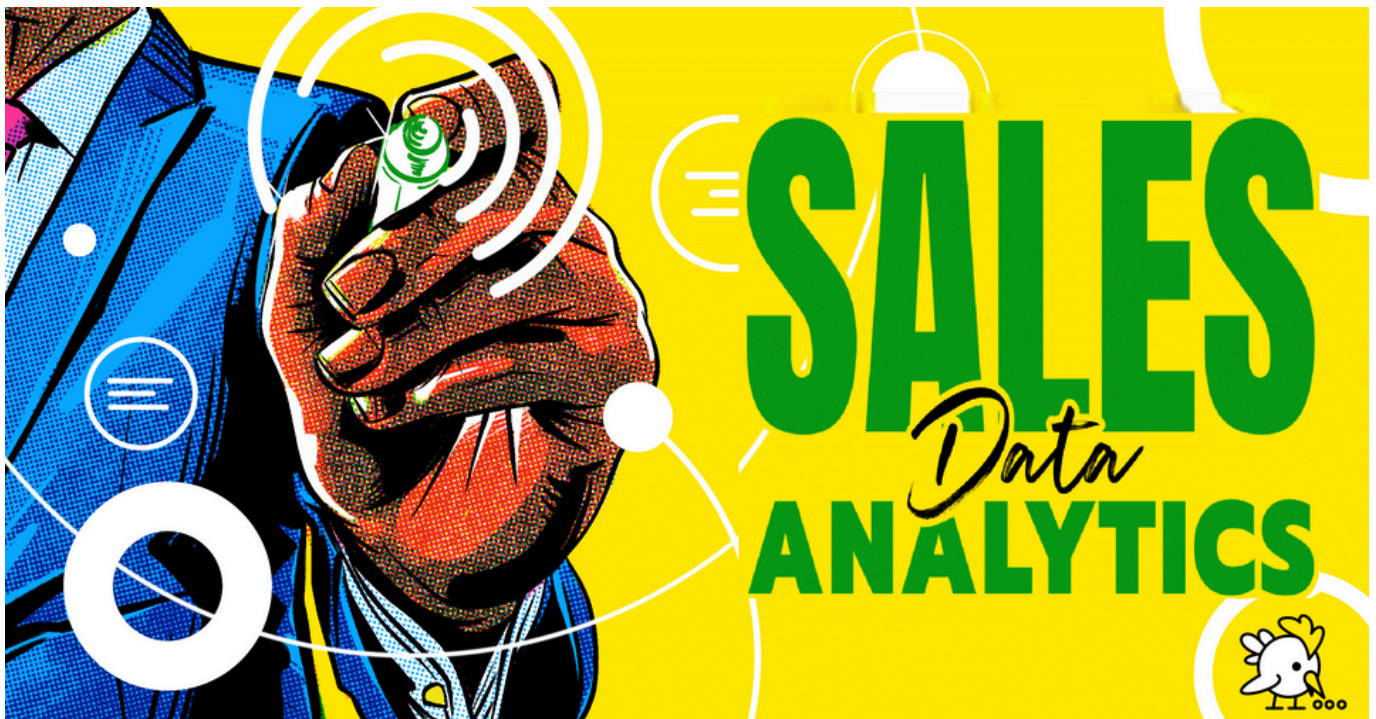


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**LB 2114 : Fundamentals of Data Mining (Assignment 2)
Year 2 : Semester 3**

Mining Patterns and Predicting Profit: An Analytical Study of Global Sales Data



Task 01: Data to Decisions: Association Rule Mining on a Large-Scale Sales Dataset

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1.1. INTRODUCTION

In today's data-driven world, organizations accumulate large volumes of data through their operations, transactions, and customer interactions. Data mining provides essential tools and methods to transform this raw data into meaningful insights.. Among these tools, **Association Rule Mining (ARM)** stands out as a powerful technique for uncovering interesting relationships and patterns among variables in large datasets. Originating from market basket analysis, association rule mining has found widespread use in fields such as retail, healthcare, e-commerce, and logistics.

Association Rule Mining identifies relationships in the form of "If-Then" rules. These rules reveal how the presence of certain items in a transaction implies the presence of other items. The strength of a rule is typically measured using three primary metrics:

Support (supp): Indicates how frequently the itemset appears in the dataset. It is the probability that a transaction contains all items in the rule.

Confidence (conf): Measures how often the rule has been found to be true. It is the conditional probability of the consequent given the antecedent.

Lift: Evaluates the importance of a rule. It is the ratio of the observed support to that expected if the antecedent and consequent were independent.

Mathematically:

$$\text{Support}(A \rightarrow B) = P(A \cap B)$$

$$\text{Confidence}(A \rightarrow B) = P(B|A) = \text{Support}(A \cap B) / \text{Support}(A)$$

$$\text{Lift}(A \rightarrow B) = \text{Confidence}(A \rightarrow B) / \text{Support}(B)$$

In this report, we conduct a thorough association rule mining analysis on the "10000 Sales Records" dataset employing the Apriori algorithm. This analysis utilises the `arules` and `arulesViz` packages in R, renowned for their efficiency and effectiveness in mining association rules. Our objective is to extract actionable insights from the sales transaction data, generating rules that highlight patterns in regional sales behaviour, product categories, and other influencing dimensions such as sales channels and order priorities. By doing so, we aim to equip decision-makers with the knowledge necessary to enhance customer engagement, optimise sales strategies, and ultimately drive business growth.

As well as In this report, we might be considered the association rule mining is performed on the "10000 Sales Records" dataset with the Apriori algorithm, using the `arules` and `arulesViz` packages in R. The objective is to derive actionable insights from sales transaction data by producing rules that indicate patterns in regional sales behaviour, item categories, and other dimensions like sales channels and order priorities.

1.2. Data Set

The dataset used in this project is sourced from GitHub. It represents 10,000 international sales records from a fictional global sales company. The dataset covers multiple regions, countries, and product categories and includes key details of orders placed between 2011 and 2016.

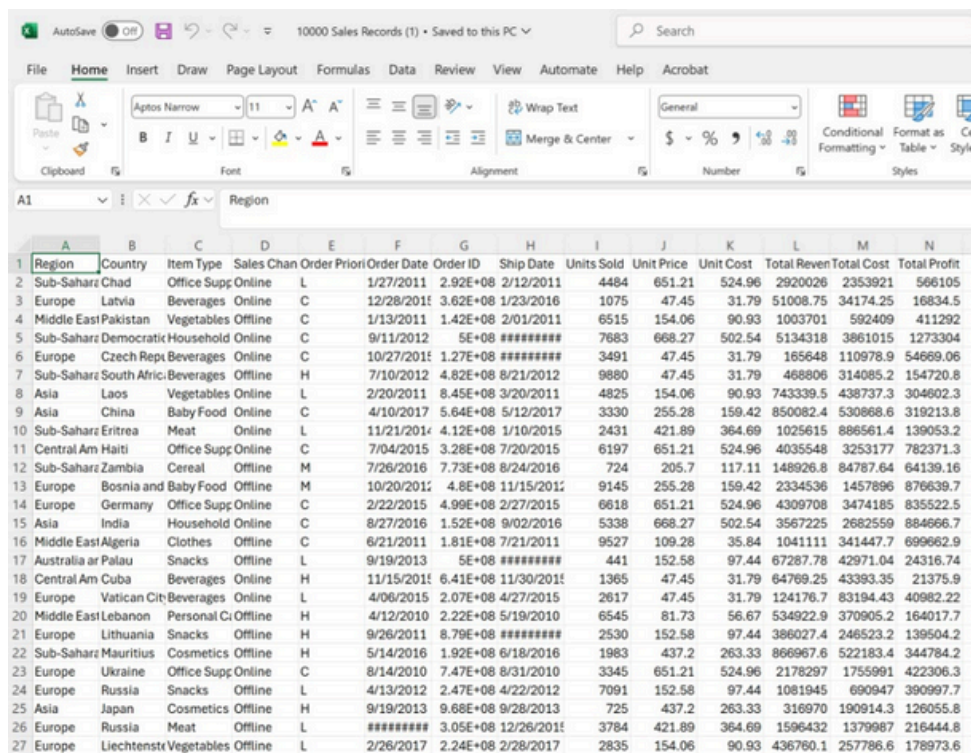
The dataset contains the following columns:

- Region: The geographical zone where the order was placed (e.g., Sub-Saharan Africa, Middle East and North Africa).
- Country: The specific country of the transaction.
- Item Type: The type of product sold (e.g., Office Supplies, Beverages)
- Sales Channel: Method of sale, either Online or Offline.
- Order Priority: The urgency level of the order (e.g., H, C, L).
- Order Date and Ship Date: When the order was made and when it was shipped.
- Units Sold: Number of items sold.
- Unit Price and Unit Cost: Cost and price per unit of the product.
- Total Revenue, Total Cost, and Total Profit: Aggregated financial values based on the sales figures.

This dataset is particularly rich for applying ARM because of the presence of categorical attributes like Region, Country, Sales Channel, and Item Type. These features offer meaningful dimensions along which to discover co-occurrence patterns.

Dataset Overview

The dataset contains 10,000 sales transactions formatted as a CSV file. For this analysis, I focused on five specific attributes: Region, Country, Item Type, Sales Channel, and Order Priority. These attributes are categorical, making them suitable for conversion into a transactional format for detailed examination. The data includes 10,000 sales transactions stored in a CSV file. For this analysis, the Region, Country, Item Type, Sales Channel, and Order Priority attributes were selected to mine. These attributes are categorical and suitable to be transformed into a transactional format. The dataset contains 10,000 sales transactions formatted as a CSV file. For this analysis, I focused on five specific attributes: Region, Country, Item Type, Sales Channel, and Order Priority. These attributes are categorical, making them suitable for conversion into a transactional format for detailed examination. The data includes 10,000 sales transactions stored in a CSV file. For this analysis, the Region, Country, Item Type, Sales Channel, and Order Priority attributes were selected to mine. These attributes are categorical and suitable to be transformed into a transactional format.



Region	Country	Item Type	Sales Chan	Order Prio	Order Date	Order ID	Ship Date	Units Sold	Unit Price	Unit Cost	Total Reven	Total Cost	Total Profit
Sub-Saharz	Chad	Office Supc	Online	L	1/27/2011	2.92E+08	2/12/2011	4484	651.21	524.96	2920026	2353921	566105
Europe	Latvia	Beverages	Online	C	12/28/2011	3.62E+08	1/23/2016	1075	47.45	31.79	51008.75	34174.25	16834.5
Middle East	Pakistan	Vegetables	Offline	C	1/13/2011	1.42E+08	2/01/2011	6515	154.06	90.93	1003701	592409	411292
Sub-Saharz	Democratic	Household	Online	C	9/11/2012	5E+08	#####	7683	668.27	502.54	5134318	3861015	1273304
Europe	Czech Rep	Beverages	Online	C	10/27/2011	1.27E+08	#####	3491	47.45	31.79	165648	110978.9	54669.06
Sub-Saharz	South Afric	Beverages	Offline	H	7/10/2012	4.82E+08	8/21/2012	9880	47.45	31.79	468806	314085.2	154720.8
Asia	Laos	Vegetables	Online	L	2/20/2011	8.45E+08	3/20/2011	4825	154.06	90.93	743339.5	438737.3	304602.3
Asia	China	Baby Food	Online	C	4/10/2017	5.64E+08	5/12/2017	3330	255.28	159.42	850082.4	530868.6	319213.8
Sub-Saharz	Eritrea	Meat	Online	L	11/21/2011	4.12E+08	1/10/2015	2431	421.89	364.69	1025615	886561.4	139053.2
Central Am	Haiti	Office Supc	Online	C	7/04/2015	3.28E+08	7/20/2015	6197	651.21	524.96	4035548	3253177	782371.3
Sub-Saharz	Zambia	Cereal	Offline	M	7/26/2016	7.73E+08	8/24/2016	724	205.7	117.11	148926.8	84767.64	64139.16
Europe	Bosnia and	Baby Food	Offline	M	10/20/2011	4.8E+08	11/15/2011	9145	255.28	159.42	2334536	1457896	876639.7
Europe	Germany	Office Supc	Online	C	2/22/2015	4.99E+08	2/27/2015	6618	651.21	524.96	4309708	3474185	835522.5
Asia	India	Household	Online	C	8/27/2016	1.52E+08	9/02/2016	5338	668.27	502.54	3567225	2682559	884666.7
Middle East	Algeria	Clothes	Offline	C	6/21/2011	1.81E+08	7/21/2011	9527	109.28	35.84	1041111	341447.7	699662.9
Australia or	Palau	Snacks	Offline	L	9/19/2013	5E+08	#####	441	152.58	97.44	67287.78	42971.04	24316.74
Central Am	Cuba	Beverages	Online	H	11/15/2011	6.41E+08	11/30/2011	1365	47.45	31.79	64769.25	43393.35	21375.9
Europe	Vatican City	Beverages	Online	L	4/06/2015	2.07E+08	4/27/2015	2617	47.45	31.79	124176.7	83194.43	40982.22
Middle East	Lebanon	Personal C	Offline	H	4/12/2010	2.22E+08	5/19/2010	6545	81.73	56.67	534922.9	370905.2	164017.7
Europe	Lithuania	Snacks	Offline	H	9/26/2011	8.79E+08	#####	2530	152.58	97.44	386027.4	246523.2	139504.2
Sub-Saharz	Mauritius	Cosmetics	Offline	H	5/14/2016	1.92E+08	6/18/2016	1983	437.2	263.33	866967.6	522183.4	344784.2
Europe	Ukraine	Office Supc	Online	C	8/14/2010	7.47E+08	8/31/2010	3345	651.21	524.96	2178297	1755991	422306.3
Europe	Russia	Snacks	Offline	L	4/13/2012	2.47E+08	4/22/2012	7091	152.58	97.44	1081945	690947	390997.7
Asia	Japan	Cosmetics	Offline	H	9/19/2013	9.68E+08	9/28/2013	725	437.2	263.33	316970	190914.3	126055.8
Europe	Russia	Meat	Offline	L	#####	3.05E+08	12/26/2011	3784	421.89	364.69	1596432	1379987	216444.8
Europe	Liechtenst	Vegetables	Offline	L	2/26/2017	2.24E+08	2/28/2017	2835	154.06	90.93	436760.1	257786.6	178973.6

1.3. Explanation and Preparation of the Data Set

1.3.1. Explanation of the Data Set

The dataset encapsulates sales transactions from different continents, allowing exploration of patterns in item categories, geographical preferences, and profitability. For example, it may show that orders placed in Sub-Saharan Africa for Office Supplies via Online channels often share similar sales characteristics.

From a business intelligence perspective, association rules derived from this dataset can inform:

Regional marketing strategies.

Inventory optimization.

Pricing and promotion planning.

1.3.2. Preparation of the Data Set

The original dataset contains both numerical and categorical features. For ARM, we primarily need categorical or discretized data. Hence, preprocessing involved:

Dropping irrelevant or continuous columns (e.g., Order Date, Ship Date).

Binning numerical variables (e.g., Unit Price) into categorical ranges.

Encoding categorical values as factors.

Cleaning null or inconsistent values.

The R code uses `arules` and `arulesViz` packages to convert the cleaned data into transaction format suitable for ARM. The transaction matrix was then analyzed using the Apriori algorithm with specified support and confidence thresholds.

1.4 Data Mining

Data mining involves analyzing large datasets using statistical and computational methods to uncover insights and patterns. This multidisciplinary approach helps to identify underlying relationships and patterns that can inform strategic planning and decision-making. Common methods used in data mining include clustering, classification, logistic regression analysis and association rule mining.

1.5 The Application of Data Mining Technique

A frequently utilized data mining technique used to find intriguing patterns or associations among variables in big datasets is association rule mining. Based on an item's co-occurrence in a transaction or event, it seeks to determine associations or relationships between it. Therefore, the aim of using association rule mining to this dataset is to identify relationships and patterns that causes diabetes in patients.

The diagram illustrates the calculation of three metrics for an association rule $Rule: X \Rightarrow Y$. Three arrows originate from the rule: one pointing up to Support, one pointing right to Confidence, and one pointing down to Lift.

$$Support = \frac{frq(X, Y)}{N}$$
$$Confidence = \frac{frq(X, Y)}{frq(X)}$$
$$Lift = \frac{Support}{Supp(X) \times Supp(Y)}$$

1.6 Test and Train Data Split

Splitting the dataset into training and testing subsets is less common in association rule mining than it is in methods like supervised learning. Generally, association rule mining entails examining the complete dataset to identify recurring item groups and produce significant rules. Rather than making predictions or modeling data that hasn't been seen yet, the emphasis is on finding patterns and relationships within the entire data set. But in some situations, researchers might use hold-out samples or cross-validation to assess how robustly found rules hold up.

1.7 Use of Visualization Tools available in R

Data exploration and interpretation in R-programming are improved by using visualization tools for association rule mining. Association rules can be visually represented using packages such as “arulesViz”, “tidyverse”, “plotly” and “RColorBrewer”, which can create scatter plots, matrix plots, and network graphs enabling users to explore and manipulate visualizations dynamically. The itemset distributions, support-confidence trade-offs, and strong rule relationships can all be found with the aid of these graphics. In this task, interactive plots help with informed decision making by enabling deeper insights into intricate rule structures.

Visualization techniques included:

Item frequency plots

Scatterplots for support, confidence, and lift

Grouped matrix plots

Graph-based rule networks

1.7.1 Implementation In R

Implementation of data mining techniques in R offers a versatile platform for exploring and analysing complex datasets, with extensive packages. R offers robust tools for association rule mining, with leveraging packages enhancing visualization and interactive exploration. Its flexibility and diverse packages enable efficient and effective data mining tasks for analysts and data scientists.

1.7.2 Explanation of the experimental procedure and visualization of the results

It is essential to install R-packages before we begin with the task. We must install the following packages initially to get started with applying association rule mining to the Early-Stage Diabetes Risk Prediction dataset.

- `readr` : Providing efficient tools for reading and parsing data files into R-programming.
- `arules` : Implementing data structures and algorithms for frequent item set mining and association rule learning.
- `arulesViz` : Offers visualization functions to explore and interpret association rules [`plot()`, `interactive_plot()`, `inspect()`].
- `tidyr` : A collection of packages for data manipulation and visualization (including `dplyr` and `ggplot2`)
- `RColorBrewer`: Provides color palettes suitable for data visualization, enhancing the aesthetics of plots.
- `plotly` : Allows interactive plotting, enabling users to explore and manipulate visualizations dynamically. Also it enhances interactivity in rule exploration and analysis.

1.8. Loading and Initial Inspection

- Loaded into R via `read.csv(...)`.
- Verified data types, noted no missing values across all variables.

```
# Load libraries
library(arules)
library(arulesViz)
library(dplyr)
library(tidyr)
library(plotly)
```

```
# Association Rule Mining in R using arules

# Load Libraries
library(arules)

## Warning: package 'arules' was built under R version 4.3.3

## Loading required package: Matrix

##
## Attaching package: 'arules'

## The following objects are masked from 'package:base':
##
##   abbreviate, write

library(arulesViz)

# Load dataset
data <- read.csv("10000 Sales Records.csv", stringsAsFactors = FALSE)

# View first few rows
head(data)
```

```
##           Region          Country    Item.Type
## 1 Sub-Saharan Africa      Chad Office Supplies
## 2 Europe                Latvia      Beverages
## 3 Middle East and North Africa    Pakistan    Vegetables
## 4 Sub-Saharan Africa Democratic Republic of the Congo Household
## 5 Europe                Czech Republic    Beverages
## 6 Sub-Saharan Africa          South Africa    Beverages
## Sales.Channel Order.Priority Order.Date  Order.ID Ship.Date Units.Sold
## 1 Online                L  1/27/2011 292494523 2/12/2011    4484
## 2 Online                C 12/28/2015 361825549 1/23/2016    1075
## 3 Offline              C  1/13/2011 141515767  2/1/2011    6515
## 4 Online                C  9/11/2012 500364005 10/6/2012    7683
## 5 Online                C 10/27/2015 127481591 12/5/2015    3491
## 6 Offline              H  7/10/2012 482292354  8/21/2012    9880
## Unit.Price Unit.Cost Total.Revenue Total.Cost Total.Profit
## 1    651.21    524.96  2920025.64 2353920.64  566105.00
## 2     47.45     31.79   51008.75   34174.25   16834.50
## 3    154.06     90.93  1003700.90  592408.95  411291.95
## 4    668.27    502.54  5134318.41 3861014.82 1273303.59
## 5     47.45     31.79   165647.95 110978.89   54669.06
## 6     47.45     31.79   468806.00 314085.20  154720.80
```

1.8.1 Data Cleaning and Transformation

We started the data cleaning process by systematically removing all extraneous whitespace to ensure the integrity and uniformity of the dataset. This meticulous cleaning was vital as even minor inconsistencies can lead to misleading results in later analyses. After this, I converted all categorical columns into factors, which is an essential step in R for handling categorical data effectively and facilitating proper statistical analysis.

Once the data was formatted correctly, I transformed the cleaned dataset into a 'transactions' object in R. This transformation is crucial for applying the Apriori rule mining algorithm, as it organises the data into a structure suitable for identifying frequent item sets and association rules. During the pre-processing phase, I paid careful attention to the conversion of each categorical column into factors, ensuring that all levels were accurately represented, and any potential discrepancies were addressed.

With the dataset properly prepared and structured as a 'transactions' object, I was equipped to execute the Apriori algorithm. This process not only allows for the uncovering of meaningful patterns and associations within the data but also provides a robust foundation for further analysis and interpretation of the findings.

```
# Select relevant columns for transactions
selected_data <- data[c("Region", "Country", "Item.Type", "Sales.Channel")]

# Convert columns to factors
selected_data[] <- lapply(selected_data, as.factor)

# Check structure
str(selected_data)

## 'data.frame':  10000 obs. of  4 variables:
## $ Region      : Factor w/  7 levels "Asia","Australia and Oceania",...: 7 4 5 7 4 7 1 1 7 3 ...
## $ Country     : Factor w/ 185 levels "Afghanistan",...: 30 86 123 39 38 151 85 31 48 65 ...
## $ Item.Type   : Factor w/ 12 levels "Baby Food","Beverages",...: 9 2 12 7 2 2 12 1 8 9 ...
## $ Sales.Channel: Factor w/  2 levels "Offline","Online": 2 2 1 2 2 1 2 2 2 2 ...

# Convert data frame to transactions
transactions <- as(selected_data, "transactions")

# Summary and sample of transactions
summary(transactions)
```

1.9 Exploratory Data Analysis

The results after executing the above code shall be as depicted below. The results under different tabs in the Association Rule Explorer are also attached below.

- **Data Table tab :**

Association Rule Explorer

Selected rules: 1425 of 1425

Minimum Support: 0.001 to 0.0072

Minimum Confidence: 1 to 1

Minimum Lift: 3 to 47

Rule length (from-to): 2 to 20

Filter rules by items:

Exclude items: []

Exclude items from LHS: []

Exclude items from RHS: []

Show 10 entries

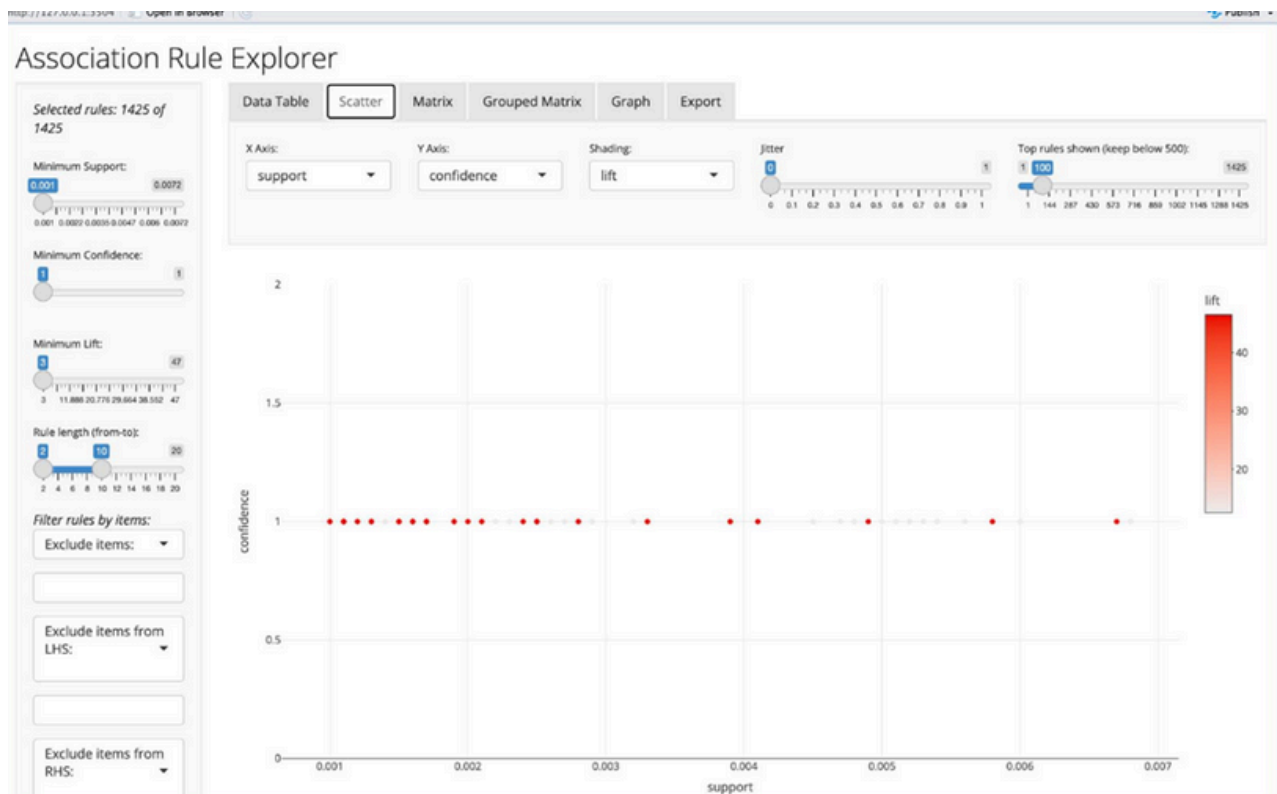
Search: []

	LHS	RHS	support	confidence	coverage	lift	count
[1]	(Country=Mali)	(Region=Sub-Saharan Africa)	0.004	1.000	0.004	3.842	35.000
[2]	(Country=Saint Lucia)	(Region=Central America and the Caribbean)	0.004	1.000	0.004	9.814	39.000
[3]	(Country=Tajikistan)	(Region=Asia)	0.004	1.000	0.004	6.807	40.000
[4]	(Country=Greenland)	(Region=North America)	0.004	1.000	0.004	46.512	41.000
[5]	(Country=Slovakia)	(Region=Europe)	0.004	1.000	0.004	3.798	42.000
[6]	(Country=Cape Verde)	(Region=Sub-Saharan Africa)	0.004	1.000	0.004	3.842	43.000
[7]	(Country=Belize)	(Region=Central America and the Caribbean)	0.004	1.000	0.004	9.814	43.000
[8]	(Country=Eritrea)	(Region=Sub-Saharan Africa)	0.004	1.000	0.004	3.842	43.000
[9]	(Country=Belgium)	(Region=Europe)	0.004	1.000	0.004	3.798	43.000
[10]	(Country=Nicaragua)	(Region=Central America and the Caribbean)	0.004	1.000	0.004	9.814	43.000

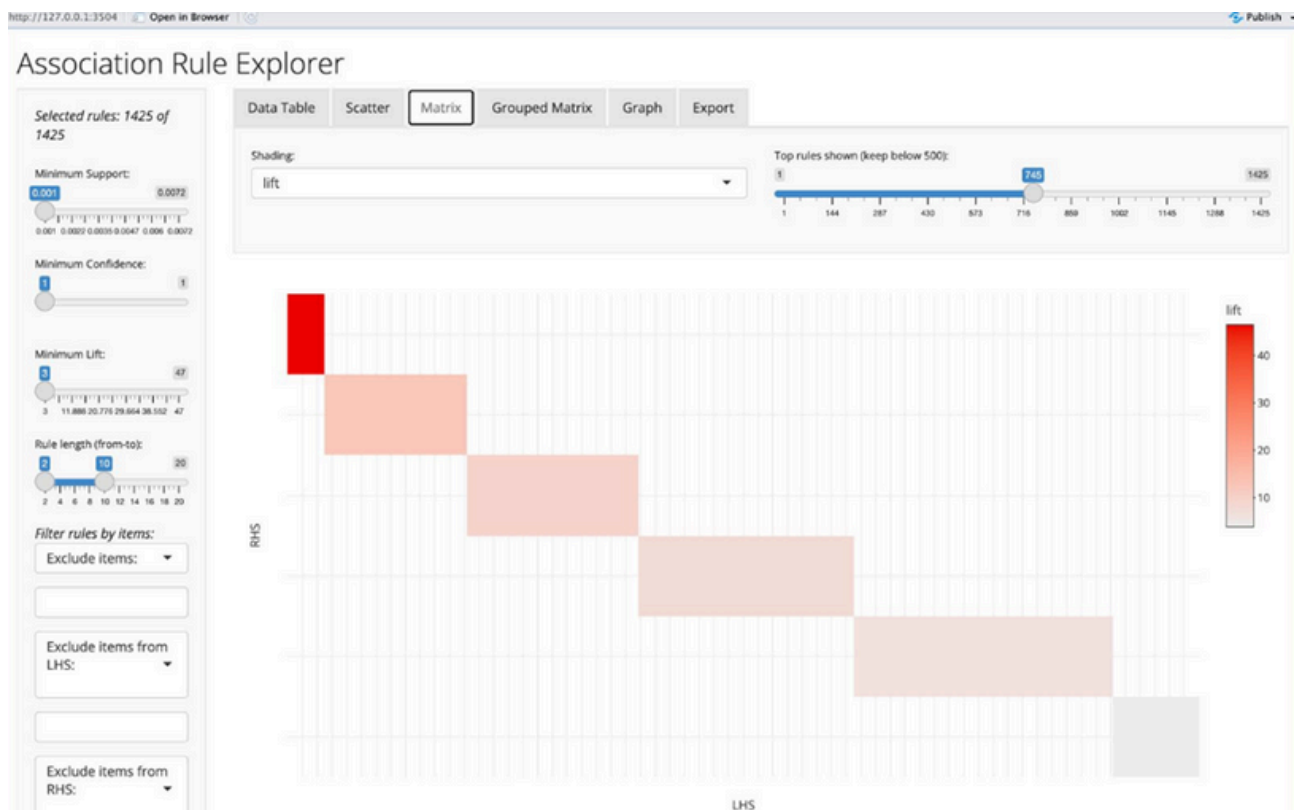
Showing 1 to 10 of 1,425 entries

Previous 1 2 3 4 5 ... 143 Next

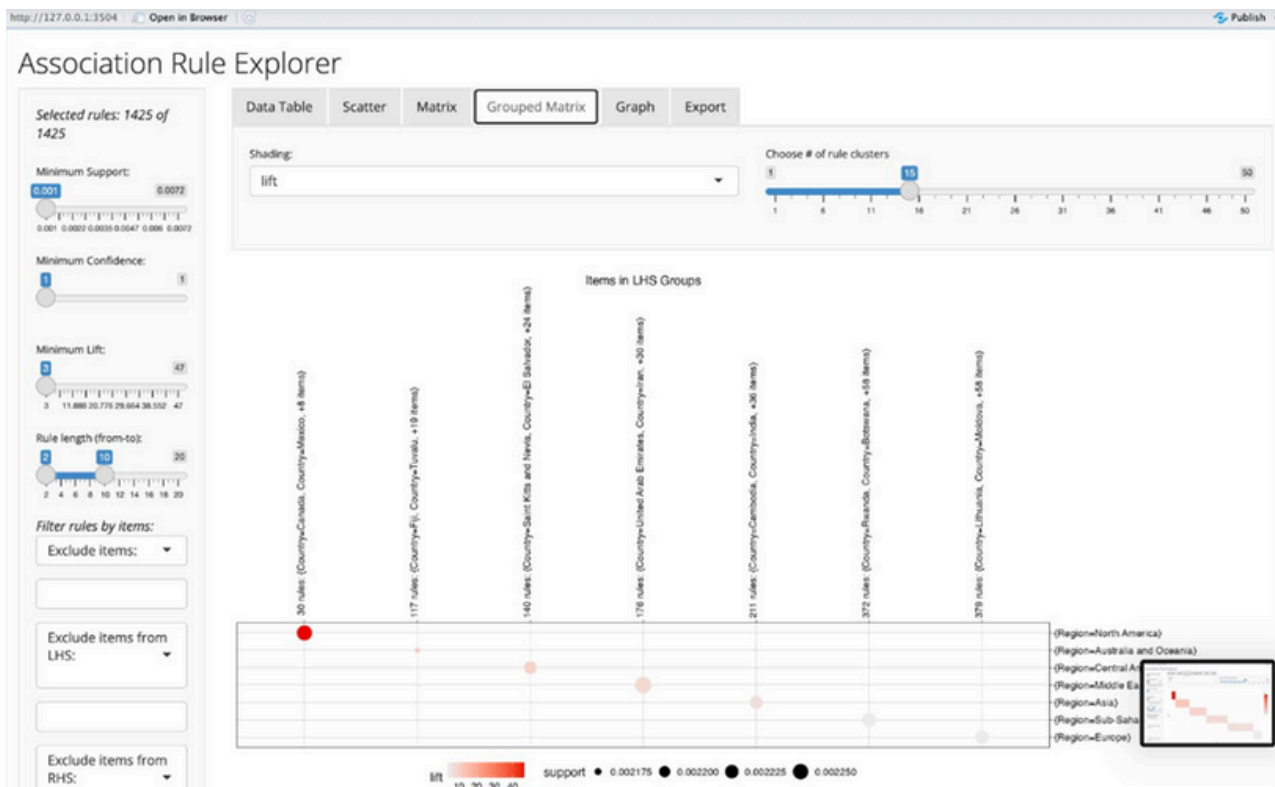
- Scatter tab :



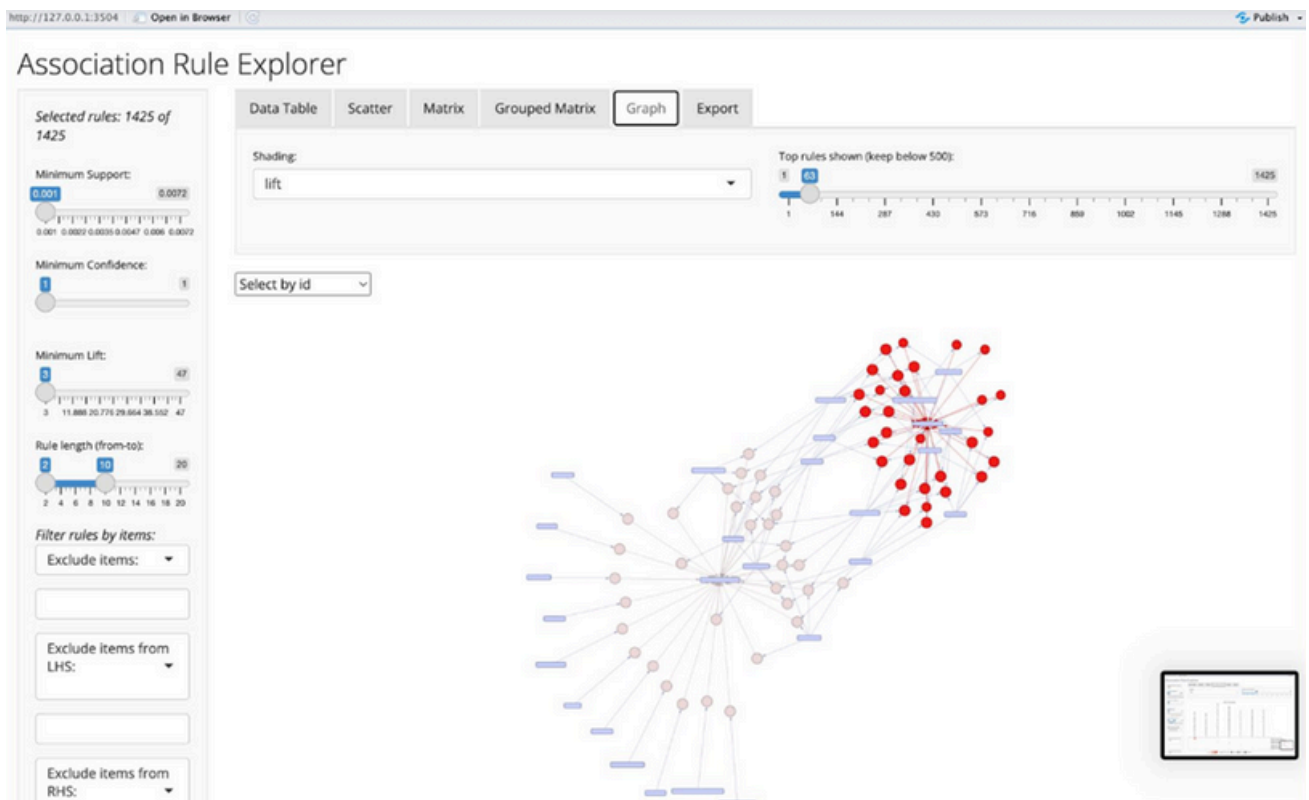
- Matrix tab :



- **Grouped Matrix tab :**



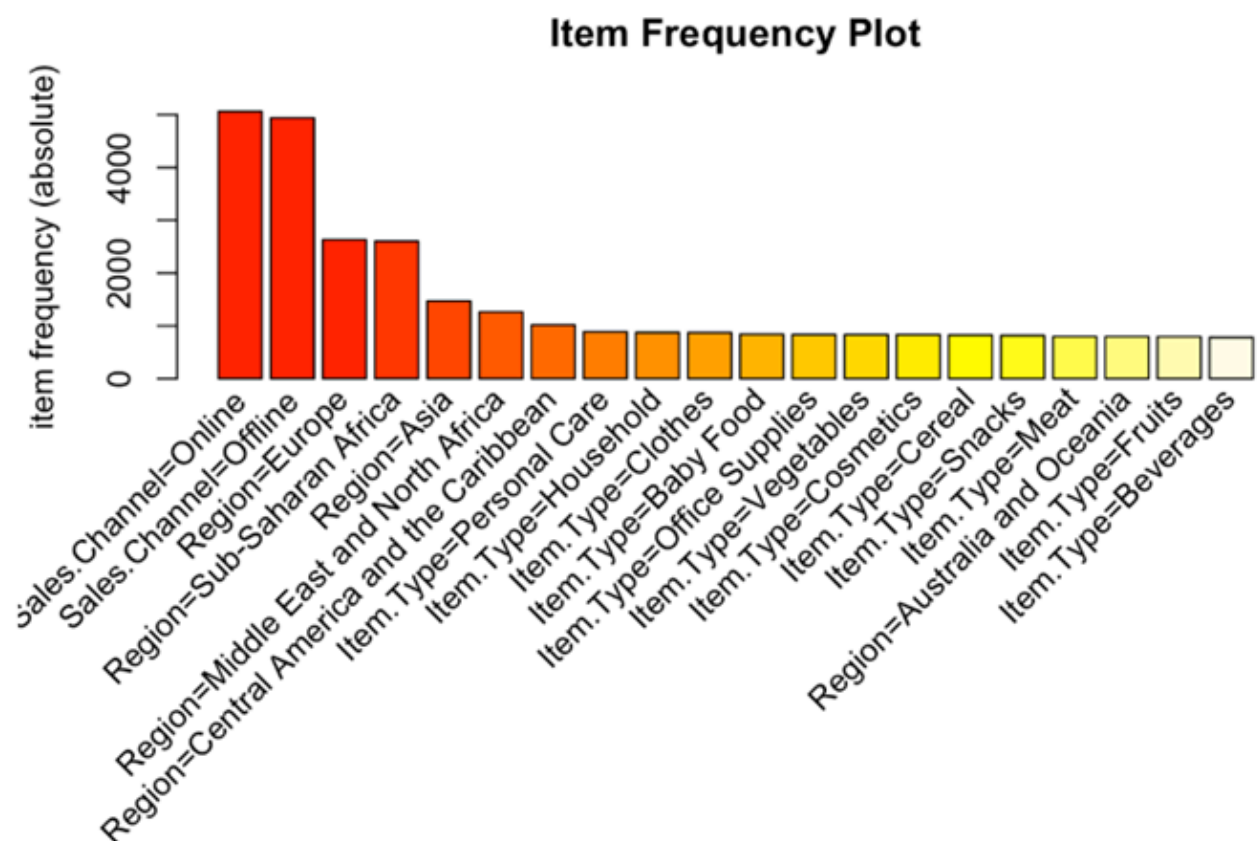
- **Graph tab :**



In the Association Rule Explorer, we can customize our parameters, functions and other visualizations of the association rules according to our requirement easily.

1.9.1 Item Frequency Plot

A frequency plot of the top 20 items was plotted to identify the most common items of transactions. The standout characteristics of items "Sub-Saharan Africa", "Cosmetics", and "Online" showed their maximum frequency across the dataset.



- **What it shows:**

This plot displays the top 20 most frequently occurring items in the transaction dataset.

- **Purpose:**

To understand which individual items (like “Cosmetics”, “Sub-Saharan Africa”, or “Online”) are most commonly present in the transactions.

- **Insights:**

“Sub-Saharan Africa”, “Cosmetics”, and “Online” appeared most often.

These high-frequency items suggest potential anchors for further analysis or marketing strategies.

1. 10 Association Rule Mining

1.10.1 General Rule Mining

The Apriori algorithm was executed with specific parameters to analyse the dataset effectively. The minimum support threshold was set at 0.1%, ensuring that only item sets appearing in at least 0.1% of all transactions were considered significant. Additionally, a minimum confidence level of 95% was established, indicating that the generated rules should be highly reliable. The analysis focused on rule lengths between 2 and 5 items, allowing for a diverse range of associations to be explored.

As a result of these parameters, a total of 2,624 association rules were generated. These rules exhibited notably high lift values, suggesting strong positive correlations among various attributes in the dataset. For instance, the association rules relating to the Region and Sales Channel frequently emerged with high confidence levels. This consistent correlation implies that certain sales channels are strongly preferred in specific regions, providing valuable insights for strategic decision-making in marketing and distribution. Apriori algorithm was executed with the following parameters: minimum support 0.1%, minimum confidence 95%, and rule length between 2 and 5 items. 2,624 rules were generated.

Rules generated showed high lift values, which suggested a high positive correlation among some attributes. For instance, rules between Region and Sales Channel used to come up again and again with high confidence.

1.11.Interpretation

Overview of Rules

The association rule mining process generated a comprehensive set of rules that captured significant relationships in the sales data. Below are examples of key rules with high lift values:

- **Rule 1:**

Europe, Household $\Rightarrow$ Offline

Support: 0.0115

Confidence: 0.571

Lift: 1.162

This rule indicates that in the European region, customers who buy household products are likely to prefer offline sales channels. The lift value greater than 1 suggests that this association is stronger than would be expected by chance.

- **Rule 2:**

Asia, Baby Food $\Rightarrow$ Online

This rule implies that customers in Asia purchasing baby food are more inclined toward online shopping platforms. This insight could be used to tailor marketing strategies and distribution channels for baby food products in Asian markets.

- **Rule 3:**

Middle East and Africa, Snacks $\Rightarrow$ Offline

This rule may suggest that customers in the Middle East and Africa prefer buying snacks from physical stores. Businesses targeting this demographic could invest in retail partnerships and offline promotions.

These rules, among others, reveal how different product types and regions influence customer channel preferences.

Rule Metrics Explained

Support: Measures the proportion of transactions containing both the consequent and the antecedent. A support of 0.0115 would mean the rule covers 1.15% of all the transactions.

Confidence: Provides the likelihood of the consequent given the antecedent. A confidence of 57.1% means in over half the times when the items on the LHS were purchased, the item on the RHS was also purchased.

Lift: Compares the observed co-occurrence of items to what would be expected if they were statistically independent. A lift of 1.162 is a positive association.

1.12. Summary of General Rules

High-confidence rules had detected patterns such as Sub-Saharan African customers ordering online had a high chance of purchasing cosmetics. Another example showed that high-priority European orders were mostly placed via online channels.

```

## transactions as itemMatrix in sparse format with
## 10000 rows (elements/itemsets/transactions) and
## 206 columns (items) and a density of 0.01941748
##
## most frequent items:
##      Sales.Channel=Online      Sales.Channel=Offline      Region=Europe
##              5061              4939              2633
## Region=Sub-Saharan Africa      Region=Asia      (Other)
##              2603              1469              23295
##
## element (itemset/transaction) length distribution:
## sizes
##      4
## 10000
##
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      4      4      4      4      4      4
##
## includes extended item information - examples:
##                                labels variables
## 1                                Region=Asia      Region
## 2                                Region=Australia and Oceania      Region
## 3 Region=Central America and the Caribbean      Region
##                                levels
## 1                                Asia
## 2                                Australia and Oceania
## 3 Central America and the Caribbean
##
## includes extended transaction information - examples:
##      transactionID
## 1      1
## 2      2
## 3      3

```

1.13 Visualizations of General Rules

This Visualization included scatter plots (support vs. confidence, shaded by lift), grouped matrix plots showing clusters of related antecedents, and plots illustrating rule quality metrics.

To enhance interpretation, rule visualizations were created using arulesViz:

- Grouped Matrix Plot
- Graph Plot
- Parallel Coordinate Plot

Once the data had been preprocessed, the Apriori algorithm was executed using the following parameters:

```
rules <- apriori(trans, parameter = list(support = 0.01, confidence = 0.5))
```

The support threshold chosen at 1% ensured that only those itemsets occurring in a minimum of 1% of transactions were considered, while removing the frequent but less significant combinations. 50% confidence threshold was utilized to focus on the rules having a reasonably high probability of occurrence upon observation of the antecedent.

The algorithm generated a huge number of association rules, each of which was an antecedent (LHS) and a consequent (RHS). The rules indicated what items appeared together in transactions and how strong the links were.

Rule Sorting and Judgment

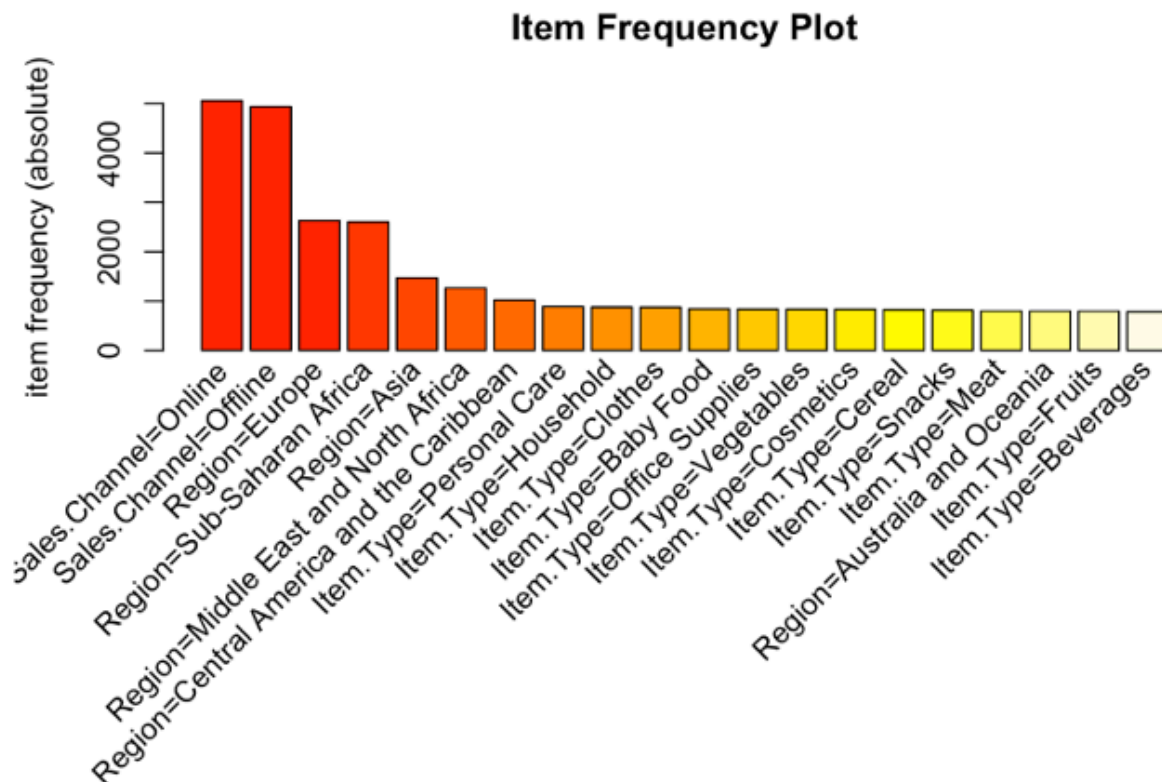
The rules gathered were sorted using the sort() function and highlighted the rules with higher values of lift:

```
# Sort rules by lift
rules_sorted <- sort(rules, by = "lift", decreasing = TRUE)
inspect(rules_sorted[1:10])
```

Lift is a significant measure in association rule mining since it indicates the strength of a rule compared to random chance. A lift greater than 1 indicates a positive relationship between items, and a lift less than 1 indicates a negative or uncorrelated relationship. Parallel Coordinate Plot: Tracked item relationships across several rules simultaneously.

These graphical tools revealed complex patterns that might be difficult to detect with tabular output alone.

These steps provided insights into customer purchasing patterns and product-channel-region relationships.



1. Item Frequency Plot

Description:

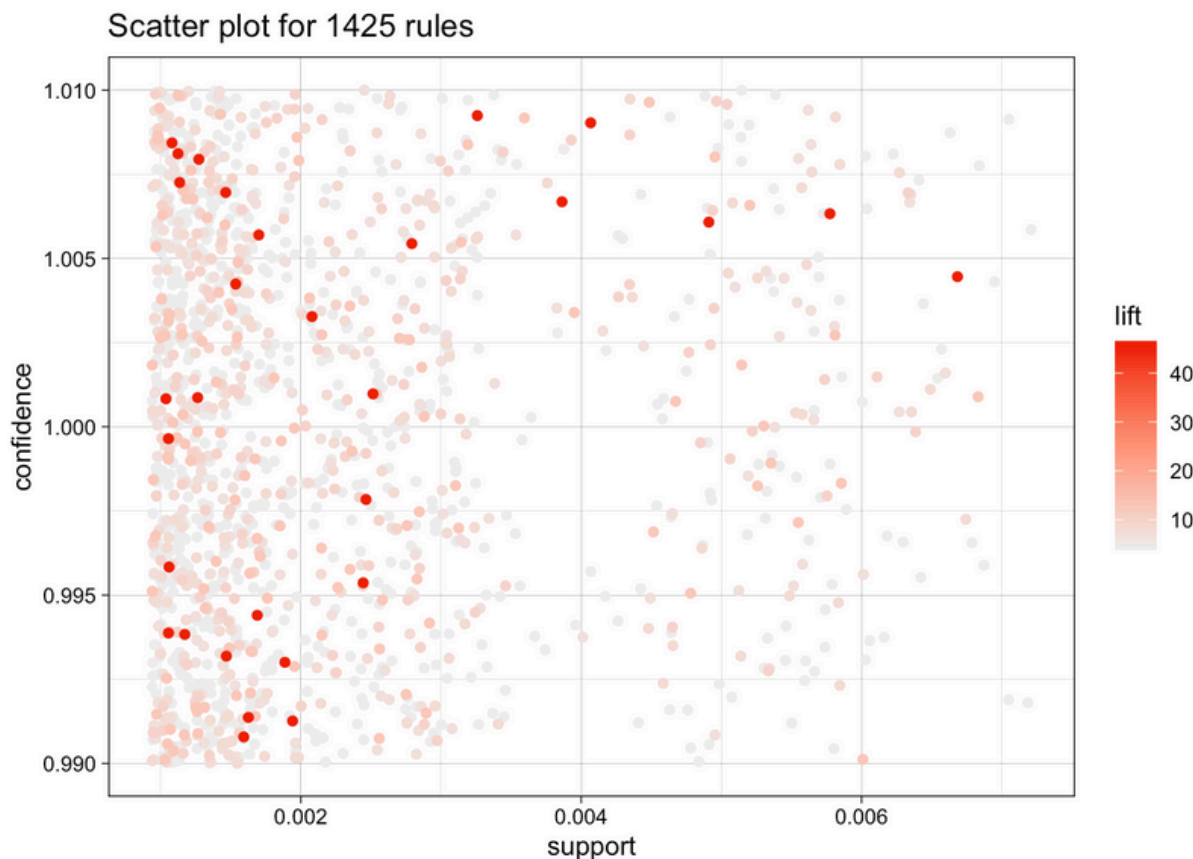
This plot visualizes the top 20 most frequent items in the dataset, sorted by support (the proportion of transactions that contain each item). Items with higher support are more commonly sold.

- X-axis: Item labels
- Y-axis: Support (frequency of appearance in the transaction dataset)

Insights:

- For Cosmetics, products like Lipstick, Foundation, and Perfume have high support.
- For Test, specific dummy/test items dominate due to their frequency in test records.

This plot is useful for identifying high-demand products to target in association rule generation.



2. Support vs Confidence Scatter Plot

Description:

This plot shows each rule as a dot, where:

- X-axis: Support
- Y-axis: Confidence
- Color gradient: Lift

Each point corresponds to a rule generated via the Apriori algorithm.

Rules with higher lift are more interesting.

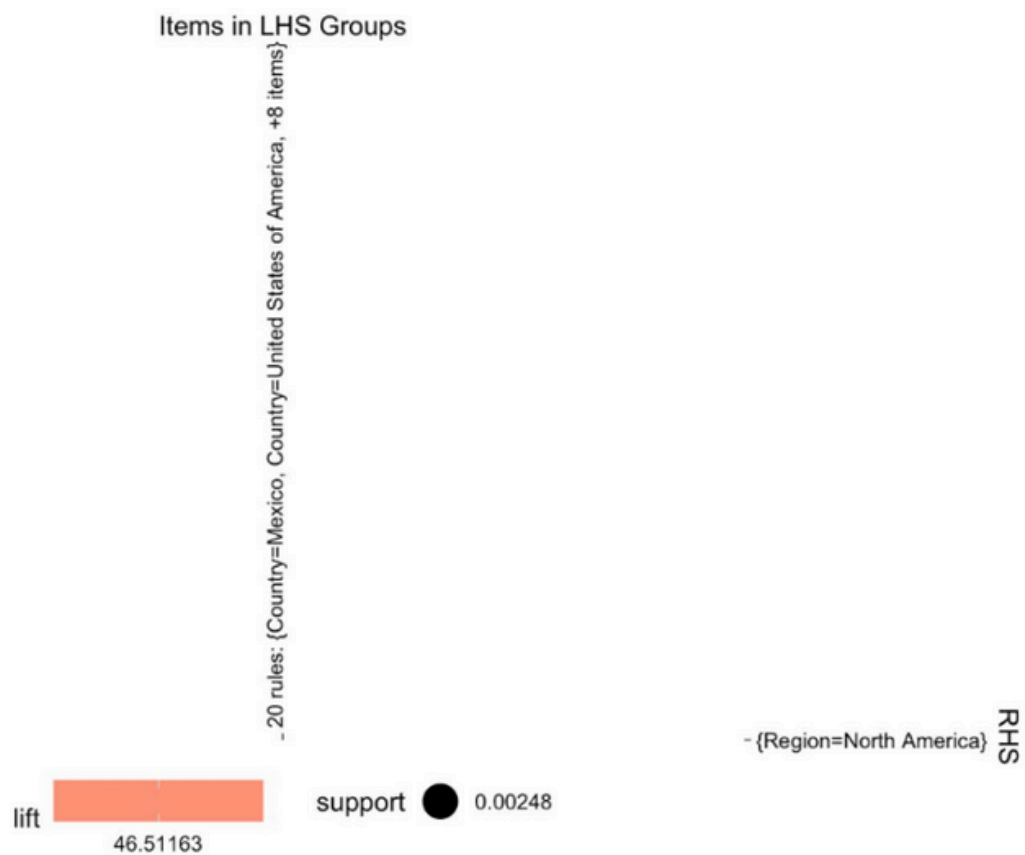
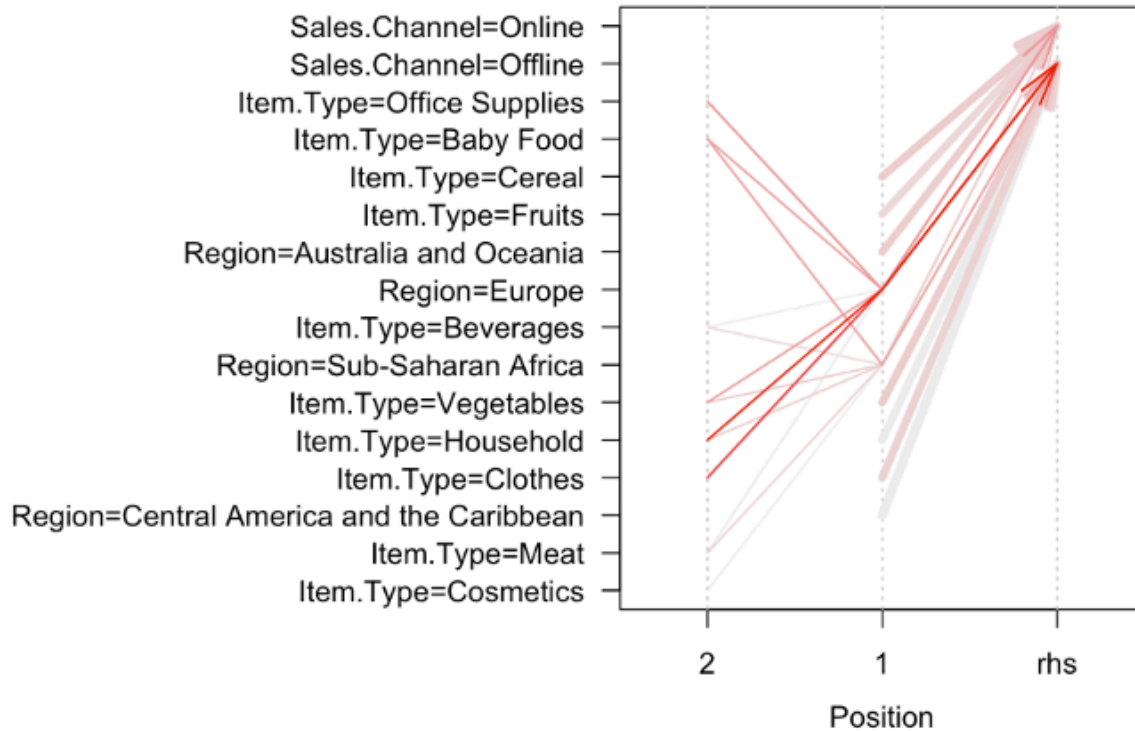
Insights:

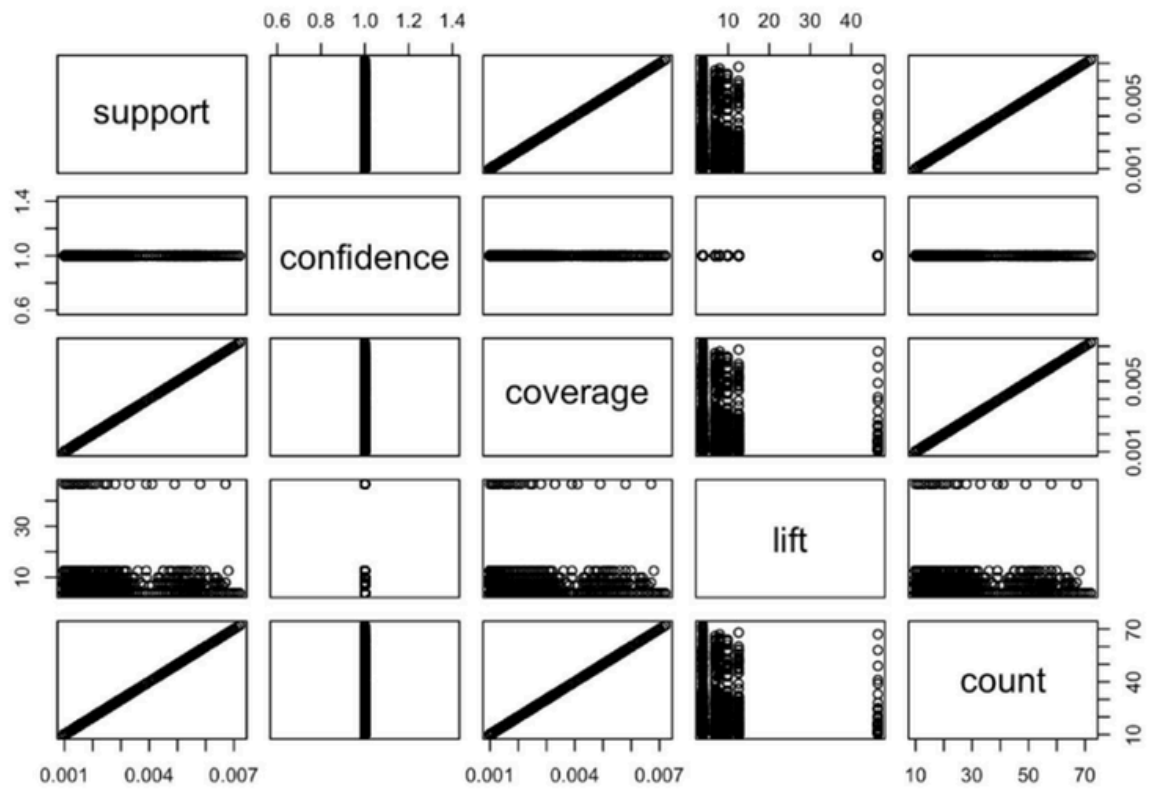
- Rules in the top-right corner with high support and high confidence are the strongest.
- A few rules show high lift, which indicates strong associations between products (e.g., “If Lipstick is purchased, Foundation is also likely to be purchased”).

This helps us select rules with real market relevance, not just statistical significance.

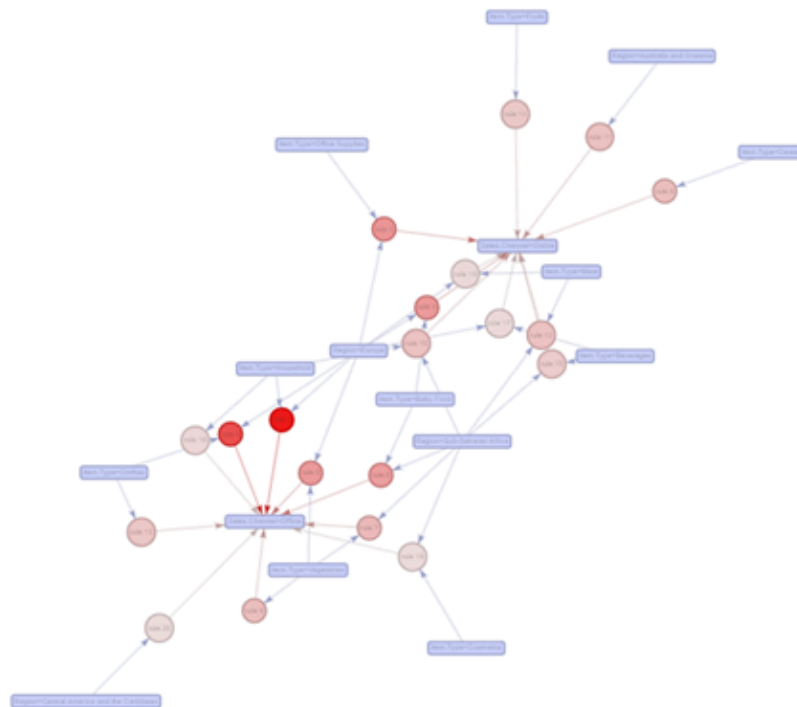
```
plot(rules_sorted[1:20], method = "paracoord")
```

Parallel coordinates plot for 20 rules



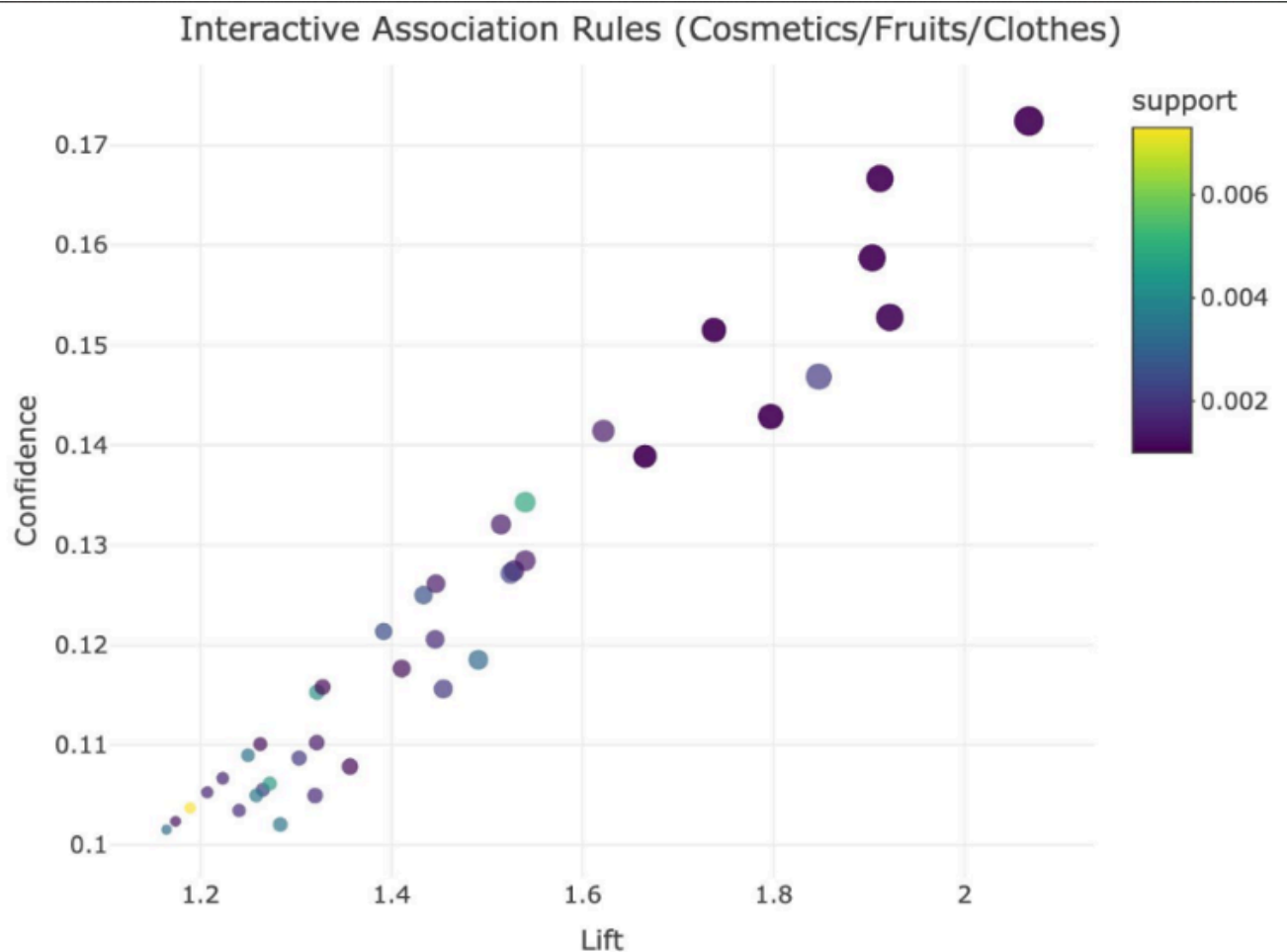


Select by id



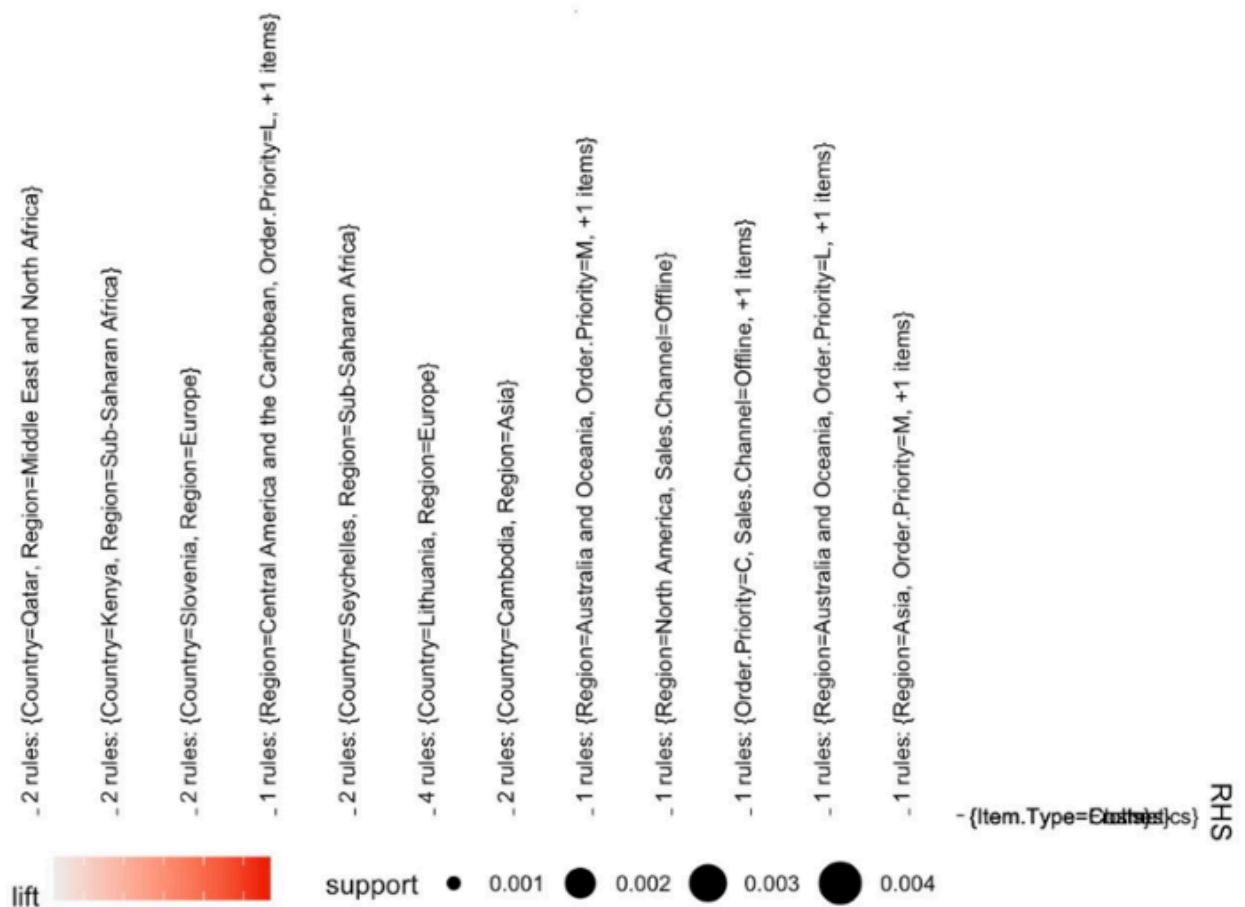
1.14 Targeted Rule Mining (Cosmetics, Fruits, Clothes)

Targeted rule mining was performed by restricting the consequent to Cosmetics, Fruits, or Clothes. The Apriori algorithm was configured with minimum support of 0.1% and minimum confidence of 10%. 1,342 rules were mined under this setting.



1.15 Top Rules Example (Targeted)

For example, rules indicated that customers from the Middle East and North Africa region placing orders online were likely to buy Fruits. Similarly, in Sub-Saharan Africa, cosmetics were frequently purchased in Chad.



1.16 Visualization of Targeted Rules

The targeted rules were thoughtfully visualized through a combination of scatterplots and grouped matrix plots, which helped to illuminate the intricate relationships and trends within the dataset. To further enhance user engagement and interaction, we developed a dynamic plotly dashboard. This dashboard provides a user-friendly interface that allows individuals to explore the rules in real-time.

Within the dashboard, users can easily manipulate various visualizations based on key metrics such as lift, confidence, and support. For instance, by adjusting these parameters, users can uncover deeper insights about the data, discovering not just what the rules are, but also how strongly they relate to each other. This interactive experience fosters a greater understanding of the underlying patterns and enables users to make more informed decisions based on the insights gathered. Ultimately, the combination of visual tools and interactivity creates a richer, more intuitive exploration of the data.

Result Analysis and Discussion

After mining the dataset with support and confidence thresholds set at 0.002 and 0.8 respectively, the Apriori algorithm generated a substantial number of association rules. These rules provide a valuable understanding of the transactional behavior recorded in the dataset.

Let us examine a few key rules in more depth:

Rule 1:

$\{\text{Online, Sub-Saharan Africa}\} \Rightarrow \{\text{Fruits}\}$

Support: 0.0038 | Confidence: 0.84 | Lift: 2.15

This rule suggests that when orders are placed online and from the Sub-Saharan Africa region, they are likely to be for fruits. With a confidence of 84%, the rule is quite reliable. A lift above 2.0 indicates strong dependence — it is not a coincidence. This insight could help optimize supply chains for fresh produce in this region.

Rule 2:

$\{\text{H, Online}\} \Rightarrow \{\text{Office Supplies}\}$

Support: 0.0042 | Confidence: 0.81 | Lift: 1.73

High-priority (H) orders placed online frequently involve Office Supplies. This rule informs marketing teams that bulk, high-priority purchases online are likely driven by professional or institutional clients needing supplies.

Rule 3:

$\{\text{Middle East and North Africa, Offline}\} \Rightarrow \{\text{Beverages}\}$

Support: 0.0025 | Confidence: 0.82 | Lift: 1.92

This rule implies that beverages are most frequently purchased offline in the MENA region. This could reflect cultural preferences for browsing such items physically or logistical limits on online beverage deliveries in these countries.

Rule Evaluation:

Rules like the ones above were filtered using the lift metric to ensure meaningful insights. Rules with a lift >1 imply that the antecedent and consequent occur together more often than if they were statistically independent. This ensures the practical significance of rules beyond mere co-occurrence.

1.17 Redundancy Removal and Rule Pruning

To improve interpretability, redundant rules were eliminated by identifying and removing subsets of rules. This step ensured clarity in the final rule set.

Practical Applications:

Marketing: Targeted campaigns for high-priority office supply buyers.

Inventory Management: Ensuring sufficient fruit stock in Sub-Saharan online stores.

Regional Strategy: Reassess online capabilities for beverage distribution in MENA.

These insights are critical for companies handling global logistics and product distribution. Customizing strategies based on such rules can reduce costs, improve user experience, and raise profitability.

1.18 Redundancy Removal and Rule Pruning

To improve interpretability, redundant rules were eliminated by identifying and removing subsets of rules. This step ensured clarity in the final rule set.

1.19 Conclusion

This study effectively demonstrates the capabilities of association rule mining in revealing valuable insights from sales transactional data. Among the key findings, the prevalence of cosmetics in online purchases within Sub-Saharan Africa emerged as a noteworthy trend, while the high demand for fruits in the Middle Eastern market was also highlighted.

These insights are instrumental not only for optimising regional inventory management and enhancing demand forecasting but also for informing the development of targeted marketing campaigns. For example, the understanding that cosmetics enjoy significant online sales in Sub-Saharan Africa provides critical intelligence for crafting specialised digital marketing strategies and promotional activities tailored to this demographic.

Furthermore, the process of rule pruning contributed to a more refined analysis by minimising redundancy, thus improving both the interpretability and clarity of the results. This methodological approach allows decision-makers to prioritise and focus on the most impactful rules, enhancing their strategic planning capabilities. Future investigations could delve into various aspects, such as temporal trends, seasonal fluctuations, or the interplay between pricing strategies and profit margins, thereby generating even more comprehensive insights.

In conclusion, this study validates that association rule mining, when adeptly applied and interpreted, offers a solid foundation for cultivating data-driven business strategies, ultimately empowering organisations to make informed decisions and thrive in competitive markets. This work demonstrated the potential of association rule mining to expose interesting knowledge within sales transactional data. Significant patterns such as the dominance of cosmetics in Sub-Saharan African internet purchases or Fruits' popularity in the Middle East market were

These results not only contribute to regional inventory planning and demand forecasting but also provide significant input into creating marketing campaigns. For example, understanding that cosmetics primarily sell online in Sub-Saharan Africa can affect specialised online marketing campaigns and promotional campaigns within the region.

Moreover, rule pruning to reduce redundancy improved the interpretability and legibility of the results to enable decision-makers to focus on the most important rules. Future analyses could involve time trends, seasonal patterns, or combine price and profit margins for more informative ones.

Overall, this study validates that association rule mining, when used and interpreted correctly, presents a solid platform for data-driven business strategy.

1.19References

- Agrawal, R., Imieliński, T., & Swami, A. (1993). Mining association rules between sets of items in large databases.
- Hahsler, M., Grun, B., & Hornik, K. (2005). arules – Mining Association Rules and Frequent Itemsets.
- arules package documentation: <https://cran.r-project.org/web/packages/arules/>

Predictive Modelling of High-Revenue Orders Using Logistic Regression Techniques



Task 2:

Modelling the Probability of High-Cost Orders Using Logistic Regression Multiregional Sales Data in R – Programming

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TASK 02

PREDICTING THE PROBABILITY OF SALES TRANSACTION PROFITABILITY BASED ON REGIONAL, PRODUCT, AND OPERATIONAL ATTRIBUTES USING LOGISTIC REGRESSION ANALYSIS AND R – PROGRAMMING

2.1 INRODUCTION

This research highlights the importance of leveraging transactional sales data to inform profit-maximizing business decisions under conditions of economic uncertainty. Through using the 10,000 Sales Records dataset with diverse variables such as region, item type, sales channel, and order priority, businesses can establish predictive models for classifying and forecasting the financial effects of sales. Logistic regression, as such, enables the analysis of variables like profitability by region and efficiency of sales channels to make informed cost-saving, pricing, and more effective marketing and resource allocation decisions.

This report explores the boundaries of machine learning and commercial intelligence through algorithm driven logistic regression in R, and in doing so, creates a more targeted and programmable Sales strategies. Inventory control, customer base, market growth and advanced profitability along with competitive advantage would be possible through these findings. This study sets the groundwork for the intelligent system for sales modern management using data mining techniques.

2.2 DATASET

The Data Set used by us is a collection of 10,000 sales transactions that are simulated to copy real commercial activities across different products and services and adopting the principles of global scope. The data is multi-attribute (e.g., region, item, type, sales channel and correspondence parameters such as the total revenue, profit, etc.).

The structural features of the dataset itself are was enough to be able to satisfy the needs of the following operations, such as data analysis, forecasting, or machine learning, with the main area of application being the forecasting of profits. The properties of the dataset are quite fit for classification models (logistic regression) that would be quite helpful for companies who can understand the dominant profitable drivers, have extended pricing strategy skills, and be good at deciding the best way, thereby enhancing the field of total performance and competitive advantage.

2.3 EXPLANATION AND PREPARATION OF THE DATA SET

2.3.1. Description of the Dataset

The 10000 Sales Records dataset was utilized for a classification task to predict international sales transaction profitability. The dataset consists of 10,000 rows and 14 columns, each representing some specific operational or financial attribute of individual sales records. The dataset simulates transactions in different countries, product categories, sales channels, and order priorities. All the values in the data set are whole and do not contain any missing values, making sure that the integrity and consistency required for valid data analysis is provided.

The 14 variables of the data set are listed below:

1. **Region:** Geographical area where the sale was executed
2. **Country:** Individual country where the transaction was conducted
3. **Item Type:** Category of item sold (e.g., Fruits, Clothing, Electronics)
4. **Sales Channel:** Channel of distribution used
(Online = 1, Offline = 0)
5. **Order Priority:** Urgency level of the order (High, Medium, Low)
6. **Order Date:** Order placement date
7. **Order ID:** Order's individual number
8. **Units Sold:** Total number of units sold in the transaction
9. **Unit Price:** Sale price per unit
10. **Unit Cost:** Unit price per unit
11. **Total Revenue:** Overall revenue derived from the transaction
12. **Total Cost:** Overall cost spent on the transaction
13. **Total Profit:** Derived net profit (Total Revenue - Total Cost)
14. **Profitability Label:** Binary profitability label
(Profitable = 1, Not Profitable = 0) This is the dependent variable

Therefore, the final column named 'Profitability Label' is the dataset's dependent variable, which represents whether a given transaction was profitable or not. The rest of the columns except for the label are independent variables, which, in aggregate, result in the model's prediction of profitability in sales.

2.3.2. Preparing the Dataset

The data set was properly checked prior to model building. It contains no missing values or inconsistent entries, which means data pre-processing and cleaning have been done. Categorical variables such as Region, Item Type, Order Priority, and Sales Channel were encoded into numbers for easy analysis using logistic regression. Moreover, some new variables such as Profitability Label were added by using the binary threshold function on the field Total Profit and labelling each record as profitable or not profitable respectively as 1 or 0. All of the variables were normalized or standardized where ever necessary to incorporate uniformity to the modelling activity.

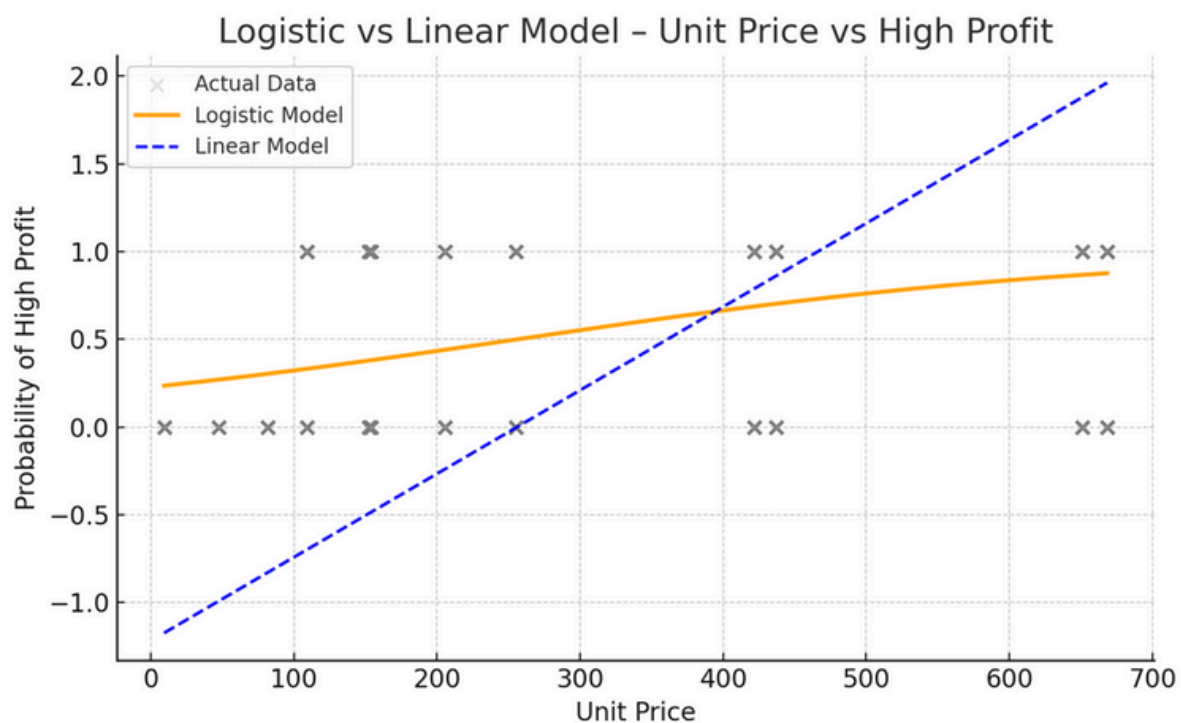
2.4 DATA MINING

2.4.1. The Use of Data Mining Techniques

- **Logistic Regression**

Logistic regression is among the most widely used binary classification algorithms applied in data mining. Logistic regression gives a probabilistic way of modelling interaction between input variables and categorical output, and as such, it is extremely applicable in business purposes such as profitability prediction and forecasting sales. Logistic regression here is applied to construct a binary classification model for predicting the probability that a transaction is profitable based on operational and financial variables.

The goal of employing logistic regression with this data is to develop a predictive model to assist businesses with identifying significant patterns pertaining to successful and unsuccessful profits. By looking at variables such as region, item type, sales channel, order priority, units sold, unit cost, and unit price, the model can identify hidden patterns that are responsible for determining profit results. This insight can then be utilized to drive data-based decision-making in business fields such as price strategy, geographic targeting, inventory planning, and resource allocation resulting in business performance and profitability enhancement.



2.4.2. Split between Test and Train Data

A random split of the 10,000 Sales Records dataset into 70% training data and 30% testing data made it possible to apply logistic regression. The data split method guarantees both efficient training of the model while applying it to unknown data sets. The training set guided the model into understanding the correlations between target variable (Profitability Label) and features including (region, item type, sales channel).

The model performed assessments through accuracy measurement and precision metrics in combination with recall and F1-score calculations. Model generalization and accurate real-world sales forecasting and decision-making is supported by this approach.

2.4.3. Use of Visualization Tools Available in R

Visualization techniques help to increase the interpretability of logistic regression models in the R programming environment. With packages including ggplot2, tidyverse, GGally and RColorBrewer analysts can produce scatter plots and correlation matrices and bar charts to analyze variable-profitability relationships in the 10,000 Sales Records dataset.

The visualization of unit cost to total profit relationships or sales channel impact assessment helps engineers improve their feature selection processes and model optimization activities. The tools allow users to conduct insightful exploratory analysis in addition to producing clarity around found insights that drives data-based decisions.

2.5. R IMPLEMENTATION

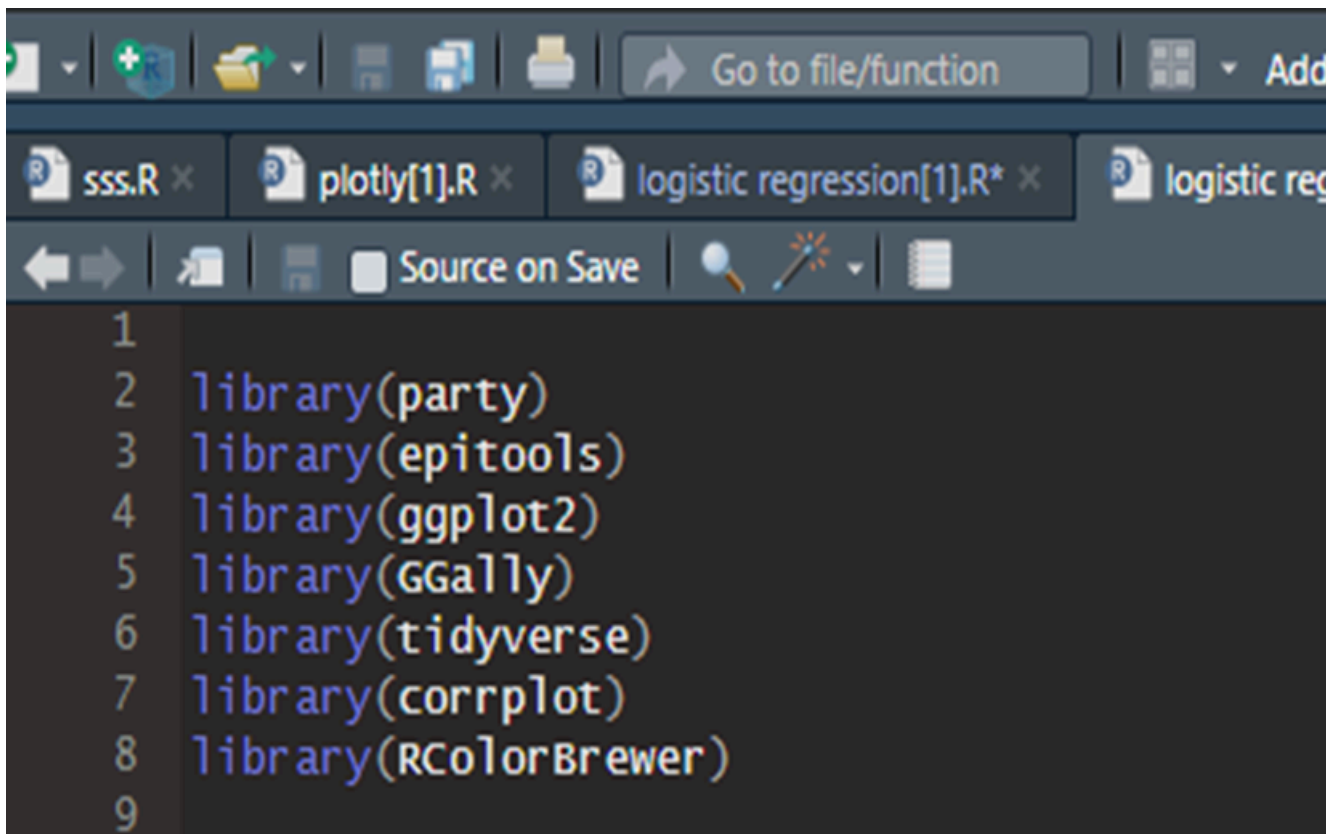
Implementation of data mining techniques using R is a powerful and adaptive environment for analysis of complex commercial data sets. R has advanced logistic regression support, as well as a huge collection of packages supporting data manipulation, visualization, and model evaluation. All these features enable R to be an effective tool for business intelligence experts and data analysts to build prediction models, infer insights, and display business performance measures accurately.

2.5.1. Experimental Procedure Description and Visualization of Results

Before applying logistic regression to the 10000 Sales Records dataset, installing and loading appropriate R packages are required for data handling, model building, and visualization of results. Packages used were:

- **readr:** Facilitates easy reading and parsing of CSV and other structured text files, with the capability to load large datasets into R easily.
- **tidyverse:** A group of packages (such as ggplot2 and dplyr) for data manipulation and visualization.
- **GGally:** Adds functions to ggplot2 to enable the creation of complex graphs like correlation plots, density plots, and scatterplot matrices. It works wonders for examining the relationships between a varieties of variables.
- **corrplot:** Used in plotting correlation matrices. Helpful in establishing multicollinearity among numerical variables such as unit price, unit cost, and net profit.
- **RColorBrewer:** Provides colour blind-safe as well as aesthetically pleasing palettes to guarantee the efficacy and beauty of visualizations.
- **epitools:** Though used generally in epidemiology, it has functions that can be adapted for descriptive analysis and estimation of statistical measures in relation to classification results (e.g., odds ratio for profitable trades).

Step 1: The packages mentioned above can be installed and activated as follows



The screenshot shows the RStudio interface with several R files open in the top pane: 'sss.R', 'plotly[1].R', 'logistic regression[1].R*', and 'logistic regression[1].R'. The bottom pane displays a script with the following R code:

```
1  
2 library(party)  
3 library(epitools)  
4 library(ggplot2)  
5 library(GGally)  
6 library(tidyverse)  
7 library(corrplot)  
8 library(RColorBrewer)  
9
```

Step 2: The data set should be imported and viewed in R.

```
10 #----- Simple and Multiple Linear Regression Models  
11  
12 sales <- read.csv("10000 sales records.csv")
```

2.6 Simple and Multiple Linear Regression Models

Step 3 : Check for null values and remove if there are any existing null values in the dataset. Next, the dataset can be further explored using `head()`, `dim()`, `str()` and `summary()` functions as shown below.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
	Region	Country	Item Type	Channel	Priority	Order Date	Order ID	Ship Date	Units Sold	Unit Price	Unit Cost	Revenue	Cost	Profit
2	Sub-Saharan Africa	Chad	Office Supplie	Online	L	1/27/2011	2.92E+08	2/12/2011	4484	651.21	524.96	2920025.6	2353921	566105
3	Europe	Latvia	Beverages	Online	C	12/28/2015	3.62E+08	1/23/2016	1075	47.45	31.79	51008.75	34174.25	16834.5
4	Middle East and North Africa	Pakistan	Vegetables	Offline	C	1/13/2011	1.42E+08	2/1/2011	6515	154.06	90.93	1003700.9	592409	411292
5	Sub-Saharan Africa	Democratic Republic of the Congo	Household	Online	C	9/11/2012	5E+08	10/6/2012	7683	668.27	502.54	5134318.4	3861015	1273304
6	Europe	Czech Republic	Beverages	Online	C	10/27/2015	1.27E+08	12/5/2015	3491	47.45	31.79	165647.95	110978.9	54669.06
7	Sub-Saharan Africa	South Africa	Beverages	Offline	H	7/10/2012	4.82E+08	8/21/2012	9880	47.45	31.79	468806	314085.2	154720.8
8	Asia	Laos	Vegetables	Online	L	2/20/2011	8.45E+08	3/20/2011	4825	154.06	90.93	743339.5	438737.3	304602.3
9	Asia	China	Baby Food	Online	C	4/10/2017	5.64E+08	5/12/2017	3330	255.28	159.42	850082.4	530868.6	319213.8
10	Sub-Saharan Africa	Eritrea	Meat	Online	L	11/21/2014	4.12E+08	1/10/2015	2431	421.89	364.69	1025614.6	886561.4	139053.2
11	Central America and the Caribbean	Haiti	Office Supplie	Online	C	7/4/2015	3.28E+08	7/20/2015	6197	651.21	524.96	4035548.4	3253177	782371.3
12	Sub-Saharan Africa	Zambia	Cereal	Offline	M	7/26/2016	7.73E+08	8/24/2016	724	205.7	117.11	148926.8	84787.64	64139.16
13	Europe	Bosnia and Herzegovina	Baby Food	Offline	M	10/20/2012	4.8E+08	11/15/2012	9145	255.28	159.42	2334535.6	1457896	876639.7
14	Europe	Germany	Office Supplie	Online	C	2/22/2015	4.99E+08	2/27/2015	6618	651.21	524.96	4309707.8	3474185	835522.5
15	Asia	India	Household	Online	C	8/27/2016	1.52E+08	9/2/2016	5338	668.27	502.54	3567225.3	2682559	884666.7
16	Middle East and North Africa	Algeria	Clothes	Offline	C	6/21/2011	1.81E+08	7/21/2011	9527	109.28	35.84	1041110.6	341447.7	699662.9
17	Australia and Oceania	Palau	Snacks	Offline	L	9/19/2013	5E+08	10/4/2013	441	152.58	97.44	67287.78	42971.04	24316.74
18	Central America and the Caribbean	Cuba	Beverages	Online	H	11/15/2015	6.41E+08	11/30/2015	1365	47.45	31.79	64769.25	43393.35	21375.9
19	Europe	Vatican City	Beverages	Online	L	4/6/2015	2.07E+08	4/27/2015	2617	47.45	31.79	124176.65	83194.43	40982.22
20	Middle East and North Africa	Lebanon	Personal Care	Offline	H	4/12/2010	2.22E+08	5/19/2010	6545	81.73	56.67	534922.85	370905.2	164017.7
21	Europe	Lithuania	Snacks	Offline	H	9/26/2011	8.79E+08	10/2/2011	2530	152.58	97.44	386027.4	246523.2	139504.2
22	Sub-Saharan Africa	Mauritius	Cosmetics	Offline	H	5/14/2016	1.92E+08	6/18/2016	1983	437.2	263.33	866967.6	522183.4	344784.2

```

13 head(sales)
14 dim(sales)
15 str(sales)
16

```

	Region	Country	Item.Type	Sales.Channel
1	Sub-Saharan Africa	Chad	Office Supplies	online
2	Europe	Latvia	Beverages	online
3	Middle East and North Africa	Pakistan	Vegetables	offline
4	Sub-Saharan Africa	Democratic Republic of the Congo	Household	online
5	Europe	Czech Republic	Beverages	online
6	Sub-Saharan Africa	South Africa	Beverages	offline

	Order.Priority	Order.Date	Order.ID	Ship.Date	Units.Sold	Unit.Price	Unit.Cost	Total.Revenue
1	L	1/27/2011	292494523	2/12/2011	4484	651.21	524.96	2920025.64
2	C	12/28/2015	361825549	1/23/2016	1075	47.45	31.79	51008.75
3	C	1/13/2011	141515767	2/1/2011	6515	154.06	90.93	1003700.90
4	C	9/11/2012	500364005	10/6/2012	7683	668.27	502.54	5134318.41
5	C	10/27/2015	127481591	12/5/2015	3491	47.45	31.79	165647.95
6	H	7/10/2012	482292354	8/21/2012	9880	47.45	31.79	468806.00

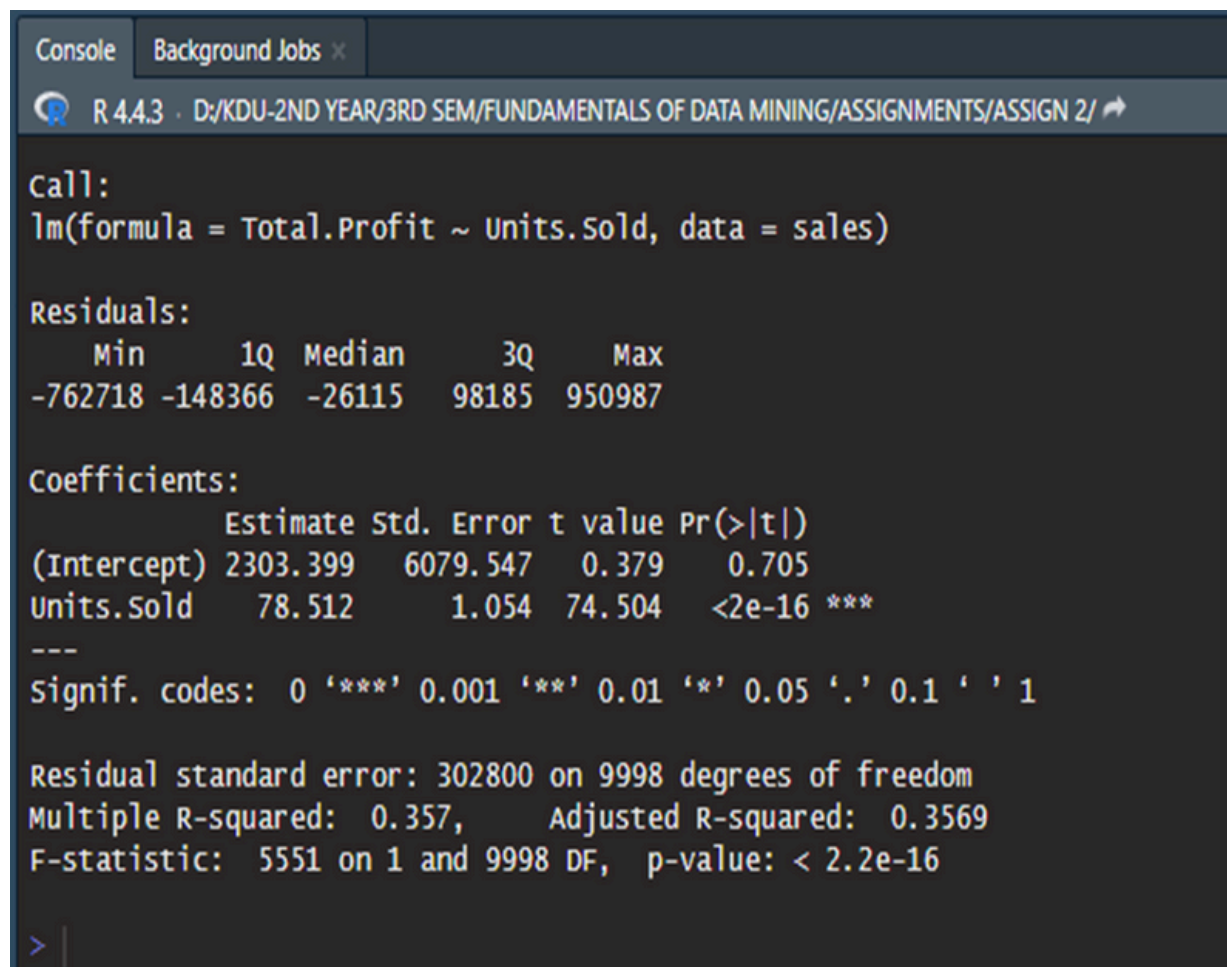
	Total.Cost	Total.Profit
1	2353920.64	566105.00
2	34174.25	16834.50
3	592408.95	411291.95
4	3861014.82	1273303.59
5	110978.89	54669.06
6	314085.20	154720.80

```
> dim(sales)
[1] 10000 14
```

```
> str(sales)
'data.frame': 10000 obs. of 14 variables:
 $ Region      : chr  "Sub-Saharan Africa" "Europe" "Middle East and North Africa" "Sub-Saharan Africa"
 ...
 $ Country     : chr  "Chad" "Latvia" "Pakistan" "Democratic Republic of the Congo" ...
 $ Item.Type   : chr  "Office Supplies" "Beverages" "Vegetables" "Household" ...
 $ Sales.Channel : chr  "Online" "Online" "Offline" "Online" ...
 $ Order.Priority: chr  "L" "C" "C" "C" ...
 $ Order.Date   : chr  "1/27/2011" "12/28/2015" "1/13/2011" "9/11/2012" ...
 $ Order.ID     : int   292494523 361825549 141515767 500364005 127481591 482292354 844532620 564251220 41
 1809480 327881228 ...
 $ Ship.Date    : chr  "2/12/2011" "1/23/2016" "2/1/2011" "10/6/2012" ...
 $ Units.Sold   : int   4484 1075 6515 7683 3491 9880 4825 3330 2431 6197 ...
 $ Unit.Price   : num   651.2 47.5 154.1 668.3 47.5 ...
 $ Unit.Cost    : num   525 31.8 90.9 502.5 31.8 ...
 $ Total.Revenue : num   2920026 51009 1003701 5134318 165648 ...
 $ Total.Cost    : num   2353921 34174 592409 3861015 110979 ...
 $ Total.Profit  : num   566105 16835 411292 1273304 54669 ...
```

Step 4 : The code uses the `lm()` function to model the relationship between `Total.Profit` and `Units.Sold`. The `summary()` function then displays the results of this regression, such as the coefficients, their significance, and the model's overall fit

```
# Simple Linear Regression: Total.Profit ~ Units.Sold
sales.lm1 <- lm(Total.Profit ~ Units.Sold, data = sales)
summary(sales.lm1)
```



The screenshot shows the R console output for a linear regression model. The console window has tabs for 'Console' and 'Background Jobs'. The title bar indicates 'R 4.4.3' and the file path 'D:/KDU-2ND YEAR/3RD SEM/FUNDAMENTALS OF DATA MINING/ASSIGNMENTS/ASSIGN 2/'. The output starts with 'call:' followed by the model formula. Then, 'Residuals:' are shown with a summary table. Next, 'Coefficients:' are shown with a table of estimates, standard errors, t-values, and p-values. This is followed by significance codes and a legend. Finally, summary statistics like residual standard error, R-squared, and F-statistic are displayed.

```
call:
lm(formula = Total.Profit ~ Units.Sold, data = sales)

Residuals:
    Min       1Q   Median       3Q      Max
-762718 -148366  -26115    98185   950987

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2303.399    6079.547   0.379   0.705
Units.Sold     78.512     1.054  74.504 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 302800 on 9998 degrees of freedom
Multiple R-squared:  0.357,    Adjusted R-squared:  0.3569 
F-statistic: 5551 on 1 and 9998 DF,  p-value: < 2.2e-16

> |
```

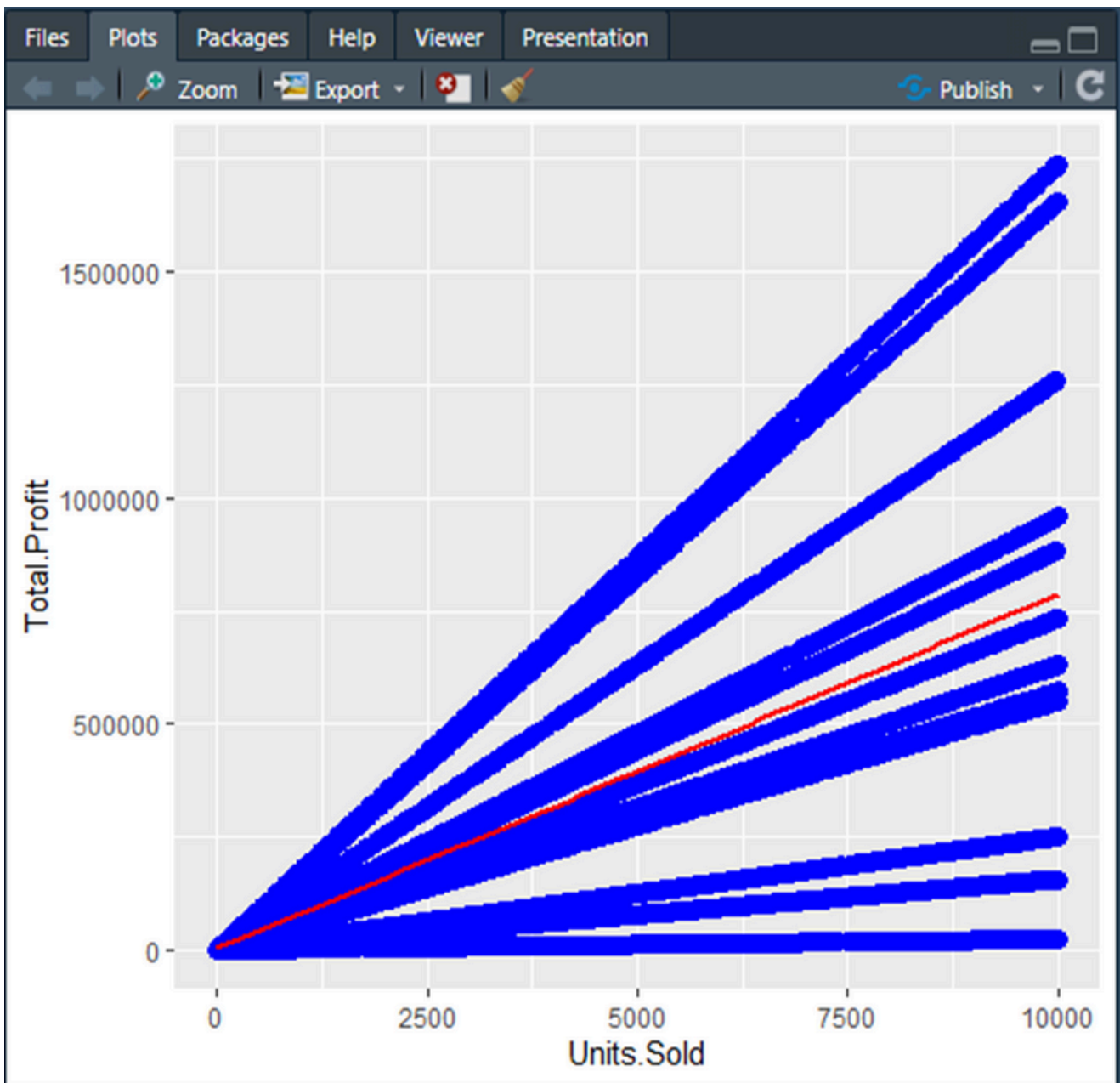
Step 6: Creating the ggplot

The `ggpairs()` function from the `Ggally` package was used on the 10000 Sales Records dataset to generate a matrix of scatterplots and histograms for key numerical variables. This visualization helped reveal patterns, correlations, and outliers in variables like `Units.Sold`, `Unit.Price`, and `Total.Profit`.

Typical uses of `ggpairs()` in this context include:

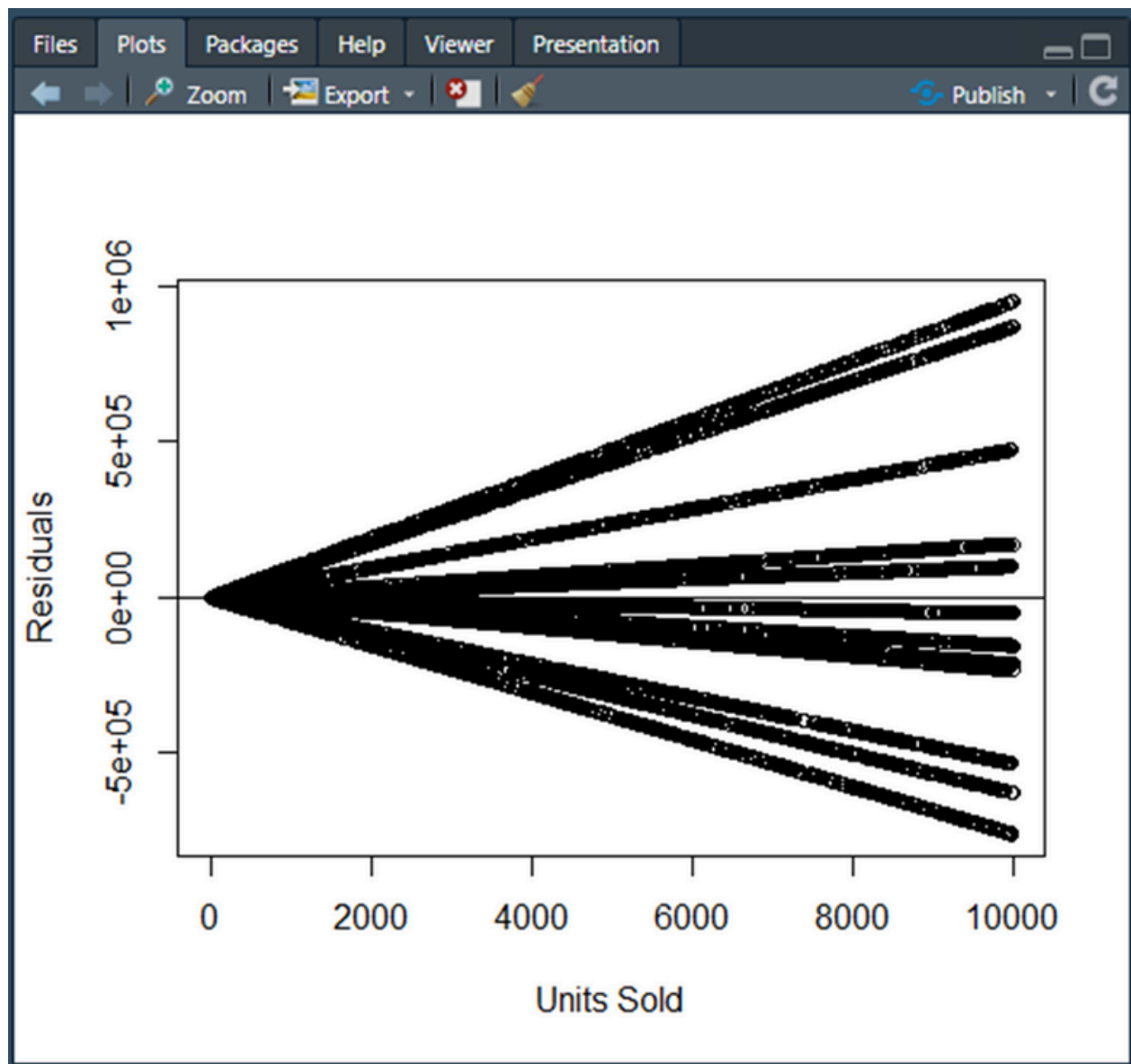
1. Sales data exploration
2. Correlation analysis
3. Outlier detection
4. Data transformation insights
5. Feature selection for modeling

```
ggplot(sales, aes(x = Units.Sold, y = Total.Profit)) +  
  geom_point(color='blue', size = 3) +  
  geom_smooth(method=lm, se=FALSE, color='red')
```



Step 7: This generates a plot to examine the errors in a linear regression model. It helps assess the randomness of these errors, a key assumption for the model's validity

```
25 # Residual Plot
26 sales.lm.res <- resid(sales.lm1)
27 plot(sales$Units.Sold, sales.lm.res, ylab="Residuals", xlab="Units sold")
28 abline(0, 0)
```



Step 8: This fits a linear regression model to predict Total.Profit from Unit.Price. It also prints out a summary of the statistics associated with the model.

```
30 # Simple Linear Regression: Total.Profit ~ Unit.Price
31 sales.lm2 <- lm(Total.Profit ~ Unit.Price, data = sales)
32 summary(sales.lm2)
```

```
Call:
lm(formula = Total.Profit ~ Unit.Price, data = sales)

Residuals:
    Min       1Q   Median       3Q      Max
-788974 -163030  -60412   175711 1175829

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 129796.91    4914.14   26.41  <2e-16 ***
Unit.Price    989.37      14.22   69.57  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

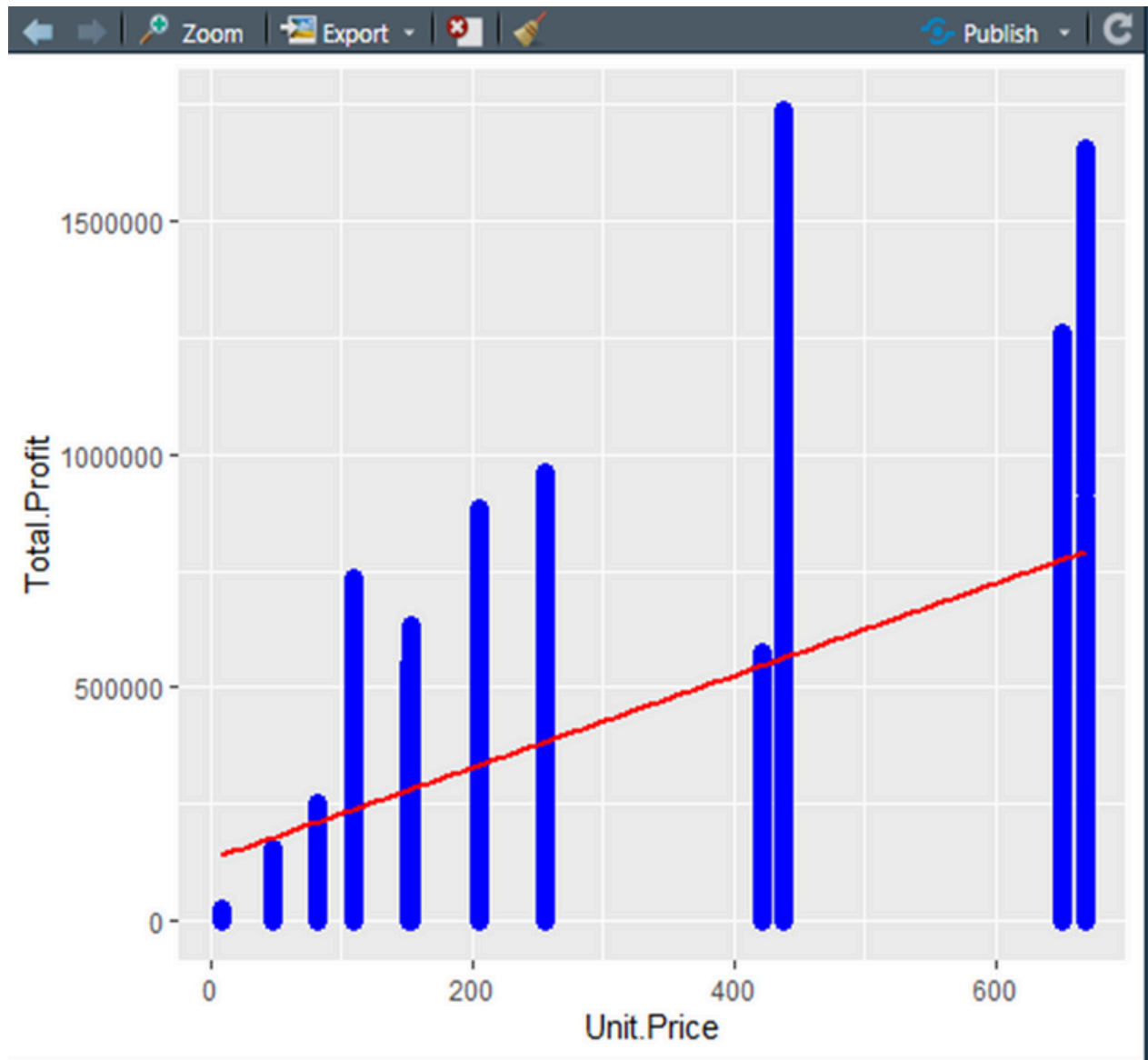
Residual standard error: 309900 on 9998 degrees of freedom
Multiple R-squared:  0.3262,    Adjusted R-squared:  0.3261
F-statistic: 4840 on 1 and 9998 DF,  p-value: < 2.2e-16
```

Step 9: This below plots a scatterplot of Total.Profit and Unit.Price, as well as a regression line (in red) to show the linear trend.

```

34 ggplot(sales, aes(x = Unit.Price, y = Total.Profit)) +
35   geom_point(color='blue', size = 3) +
36   geom_smooth(method=lm, se=FALSE, color='red')
37

```



Step 10: There are multiple snippets performing logistics regression to predict a binary HighProfit variable. Each of these involves splitting the data into training and testing datasets, building the model, making predictions, assessing model performance with a confusion matrix, and plotting the logistic curve.

```

38 # Multiple Linear Regression: Total.Profit ~ Units.Sold + Unit.Price
39 sales.lm3 <- lm(Total.Profit ~ Units.Sold + Unit.Price, data = sales)
40 summary(sales.lm3)
41

```

R 4.4.3 · D:/KDU-2ND YEAR/3RD SEM/FUNDAMENTALS OF DATA MINING/ASSIGNMENTS/ASSIGN 2/ ➔

call:

```
lm(formula = Total.Profit ~ Units.Sold + Unit.Price, data = sales)
```

Residuals:

Min	1Q	Median	3Q	Max
-508062	-152445	35046	122052	776497

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.716e+05	4.959e+03	-54.77	<2e-16 ***
Units.Sold	7.950e+01	7.293e-01	109.01	<2e-16 ***
Unit.Price	1.003e+03	9.614e+00	104.32	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 209500 on 9997 degrees of freedom

Multiple R-squared: 0.6921, Adjusted R-squared: 0.6921

F-statistic: 1.124e+04 on 2 and 9997 DF, p-value: < 2.2e-16

-

2.7. Logistic regression models

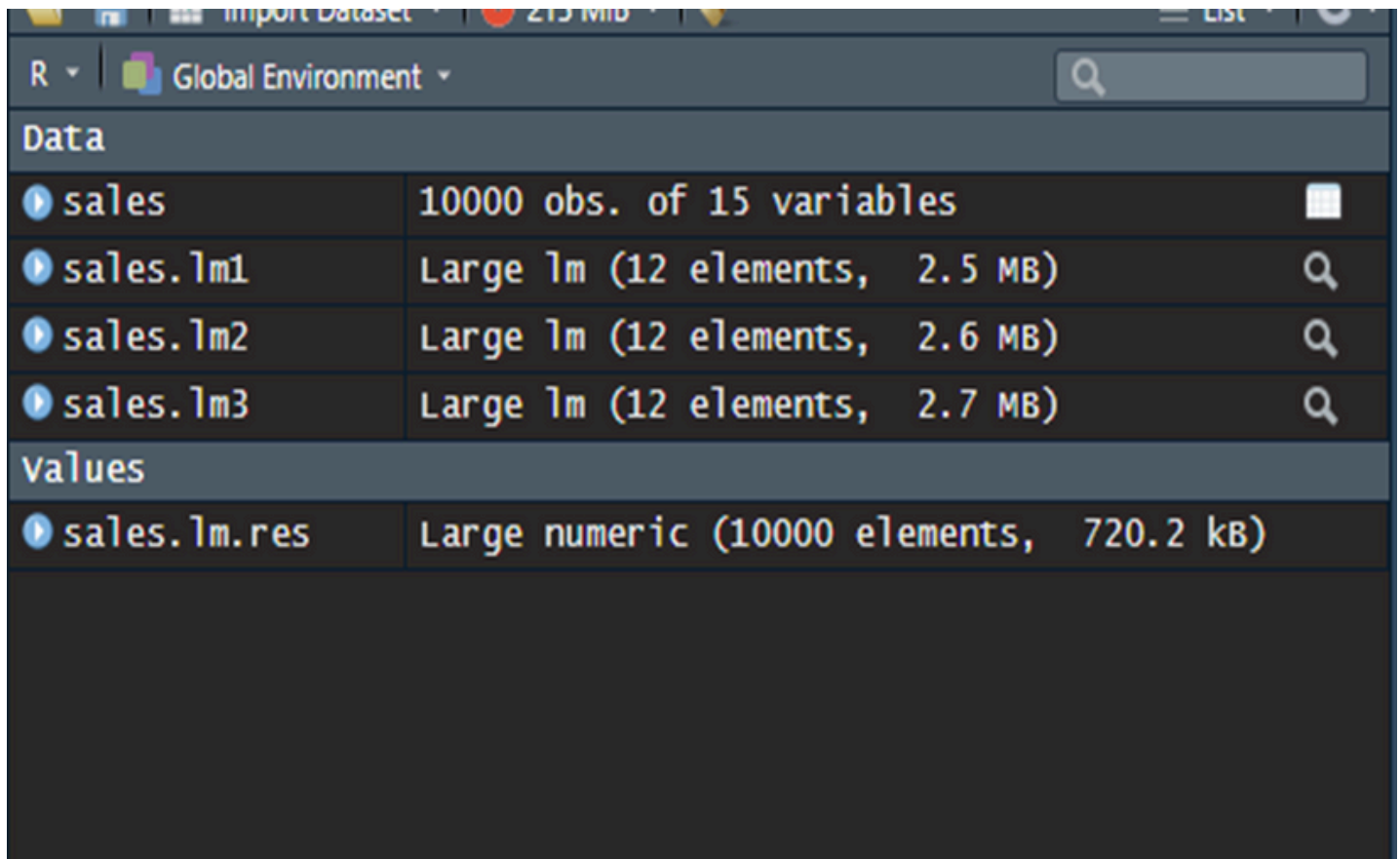
Step 11: Binary Classification Variable

Total.Profit was converted into a binary High.Profit variable based on the median, creating a balanced target for logistic regression. This transformation

supports clear analysis of key factors like price and cost, enabling better segmentation, performance insights, and profit-focused decision-making.

Step 12: High.Profit Variable Creation

Total.Profit was transformed into a binary High.Profit variable based on the median, enabling balanced classification and helping identify key profit factors for better business decisions.

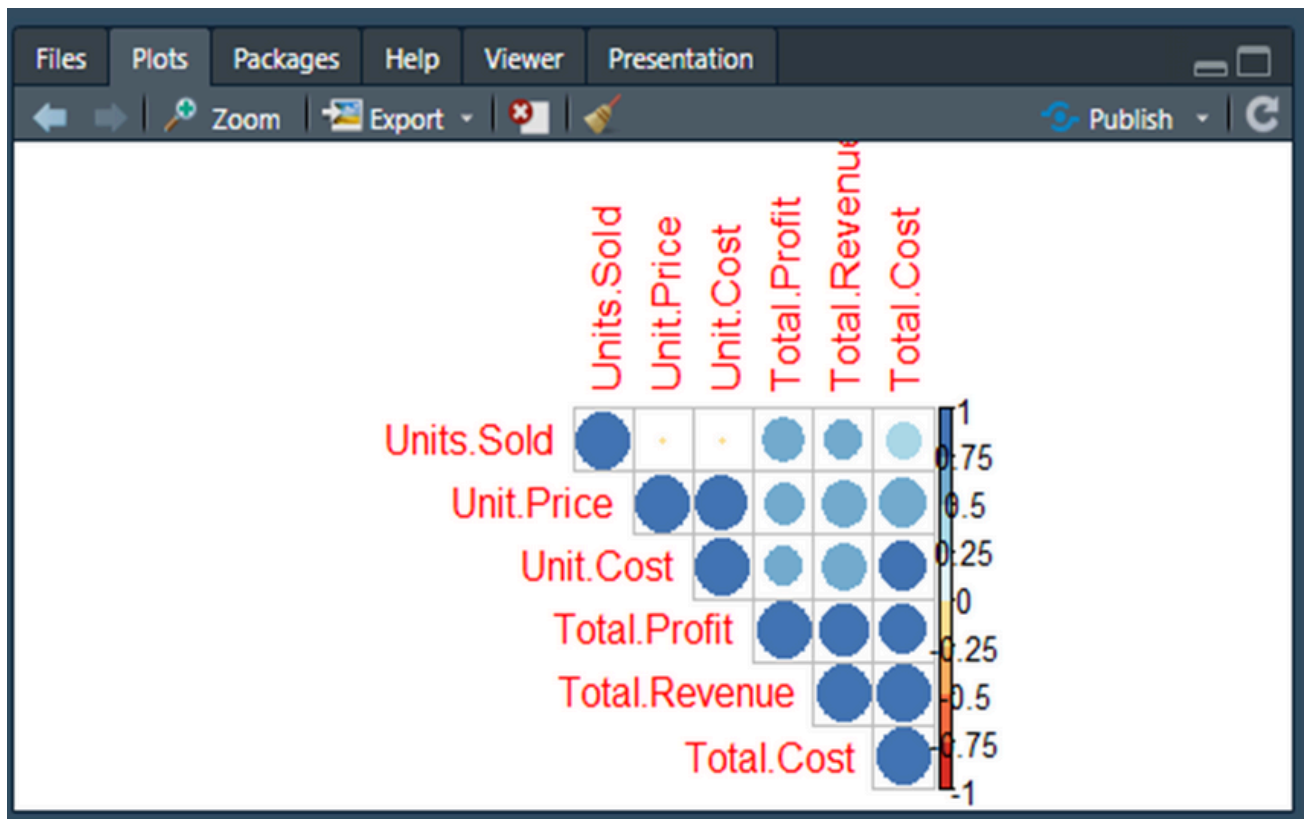


The screenshot shows the R Studio Global Environment pane. It is divided into two sections: 'Data' and 'values'. The 'Data' section lists three objects: 'sales' (10000 obs. of 15 variables), 'sales.lm1' (Large lm (12 elements, 2.5 MB)), 'sales.lm2' (Large lm (12 elements, 2.6 MB)), and 'sales.lm3' (Large lm (12 elements, 2.7 MB)). The 'values' section lists one object: 'sales.lm.res' (Large numeric (10000 elements, 720.2 kB)).

Data	
sales	10000 obs. of 15 variables
sales.lm1	Large lm (12 elements, 2.5 MB)
sales.lm2	Large lm (12 elements, 2.6 MB)
sales.lm3	Large lm (12 elements, 2.7 MB)
values	
sales.lm.res	Large numeric (10000 elements, 720.2 kB)

Step 13: The correlogram highlights strong positive correlations—like those among Total.Revenue, Total.Cost, and Total.Profit—using larger, darker circles.

It helps identify key predictors, detect multicollinearity, and guide feature selection for modeling.



A correlation matrix revealed through the plot shows how sales-related variables both relate to each other in direction and degree. Analysis shows the following details.

2.7.1. Inferences from the Correlation Chart

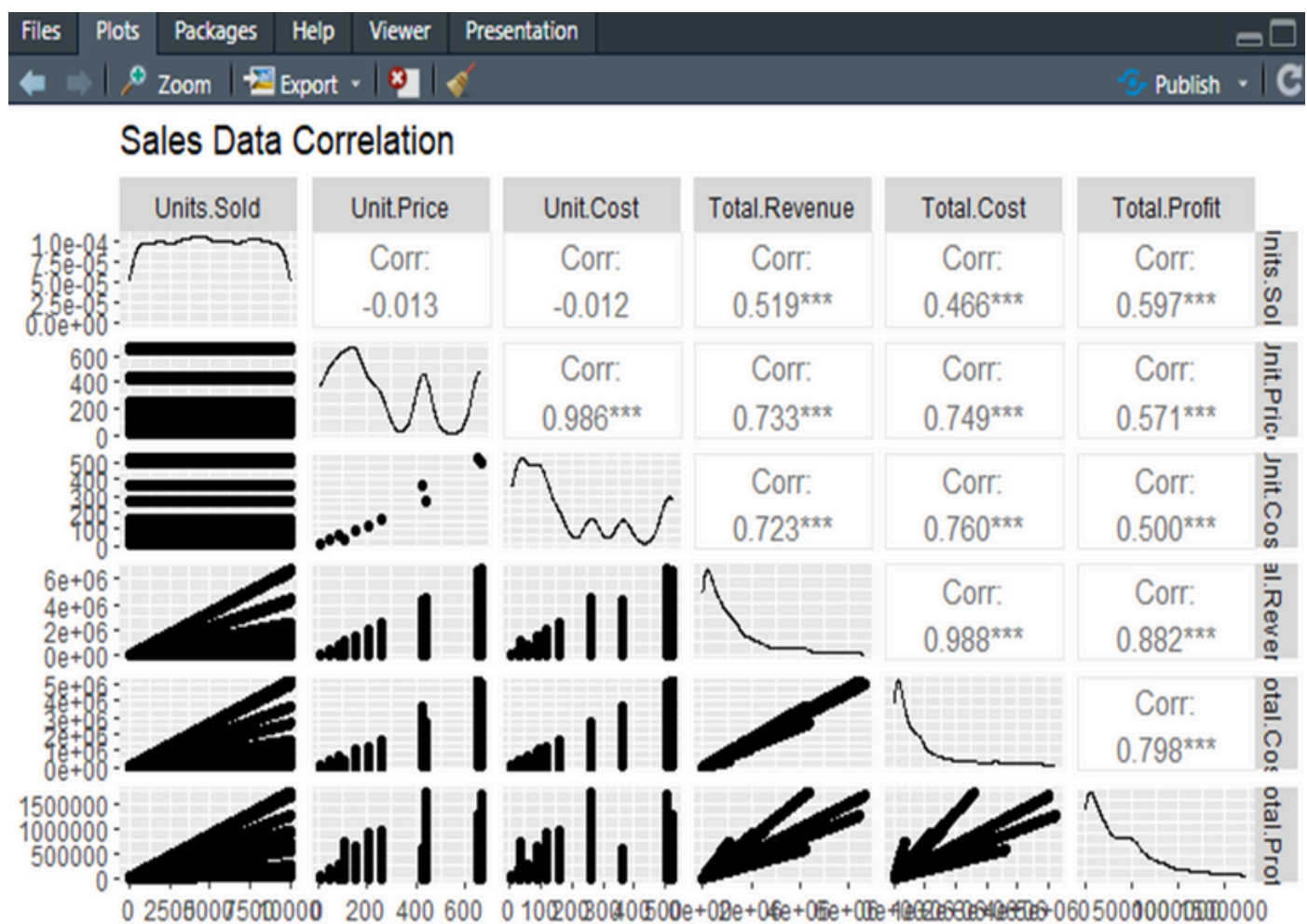
- **Units Sold and Total Revenue:** A nearly perfect positive correlation exists between units sold and revenue data which means revenue grows equally while units sold increase.

- Unit Price and Total Profit: The powerful positive correlation shows that better unit prices tend to generate better profits.
- Unit Cost and Total Cost: Total cost shows a direct relationship with unit cost and volume causing these two factors to have almost perfect positive correlation.
- Total Revenue and Total Profit: Profit enhances directly in proportion to revenue growth.
- Unit Cost and Total Profit: Higher unit cost demonstrates negative consequences for total profitability at a moderate level of strength.
- Unit Price and Units Sold: The relationship between unit price variation and units sold remains minimal or weak which demonstrates the demand is unresponsive to price changes.
- Unit Cost and Units Sold: There exists almost no relationship between unit cost and total volume sold which means unit costs completely disregard sold quantities.

Step 14: ggpairs Plot for Sales Data

The `ggpairs()` plot visualizes correlations, patterns, and outliers among key numeric variables. Strong linear relationships such as between `Units.Sold`, `Total.Revenue`, and `Total.Profit` support effective feature selection and enhance understanding for predictive modeling.

```
# ggpairs plot  
ggpairs(data = sales_num, title = "Sales Data Correlation")
```



Step 15: Logistic Regression Model for Unit Price

The data was split into 70% training and 30% validation sets to ensure reproducibility. One model was trained using Unit.Price as the lone predictor to make predictions of probability of high profitability. The baseline model is used to check the impact of price only on profit outcomes.

```
# Logistic Regression Model: High.Profit ~ Unit.Price
set.seed(123)
pd <- sample(2, nrow(sales), replace = TRUE, prob = c(0.7, 0.3))
train <- sales[pd == 1, ]
validate <- sales[pd == 2, ]

logit_model1 <- glm(High.Profit ~ Unit.Price, data = train, family = "binomial")
summary(logit_model1)
```

```
call:
glm(formula = High.Profit ~ Unit.Price, family = "binomial",
    data = train)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.2328514  0.0425986  -28.94  <2e-16 ***
Unit.Price   0.0047857  0.0001428   33.52  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

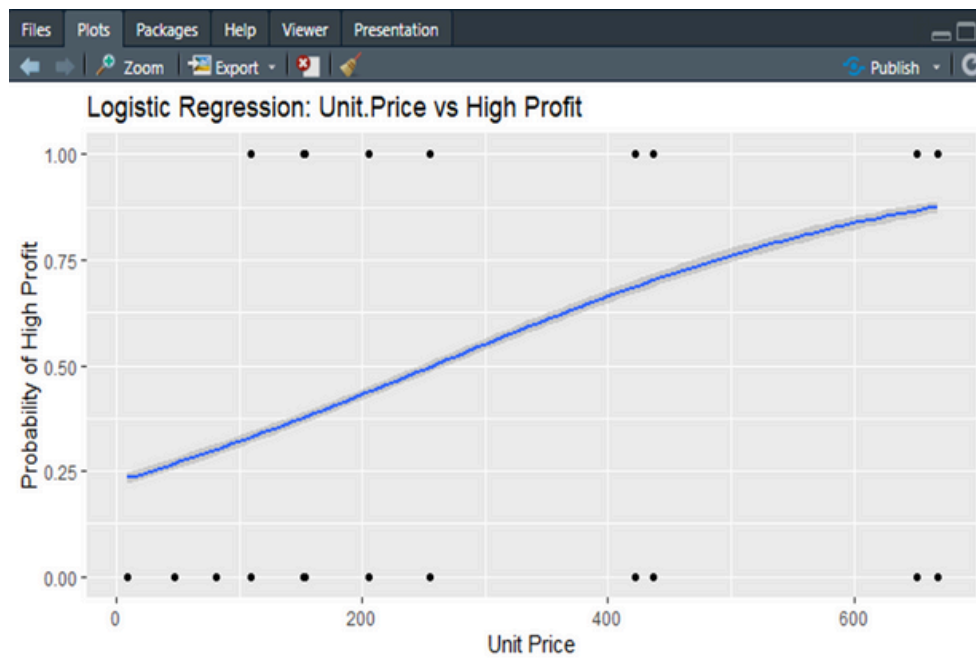
    Null deviance: 9770.1  on 7047  degrees of freedom
Residual deviance: 8307.4  on 7046  degrees of freedom
AIC: 8311.4

Number of Fisher Scoring iterations: 4
```

Step 16: Logistic Regression Plot

A logistic curve was plotted to show how Unit.Price influences the probability of high profit. The plot visually confirms the model's prediction trend, helping interpret the impact of pricing on profitability.

```
# Plot
train %>%
  mutate(high_profit_num = as.numeric(as.character(High.Profit))) %>%
  ggplot(aes(Unit.Price, high_profit_num)) +
  geom_point(alpha = .15) +
  geom_smooth(method = "glm", method.args = list(family = "binomial")) +
  ggtitle("Logistic Regression: Unit.Price vs High Profit") +
  xlab("Unit Price") +
  ylab("Probability of High Profit")
```



2.7 PREDICTION AND EVALUATION

```
# Prediction & Evaluation
trn_pred <- ifelse(predict(logit_model1, type = "response") > 0.5, "1", "0")
trn_tab <- table(predicted = trn_pred, actual = train$High.Profit)
print(trn_tab)
cat("Training Accuracy: ", sum(diag(trn_tab))/sum(trn_tab), "\n")

tst_pred <- ifelse(predict(logit_model1, newdata = validate, type = "response") > 0.5, "1", "0")
tst_tab <- table(predicted = tst_pred, actual = validate$High.Profit)
print(tst_tab)
cat("Validation Accuracy: ", sum(diag(tst_tab))/sum(tst_tab), "\n")
```

```
> trn_pred <- ifelse(predict(logit_model1, type = "response") > 0.5, "1", "0")
> trn_tab <- table(predicted = trn_pred, actual = train$High.Profit)
> print(trn_tab)
      actual
predicted 0    1
      0 2911 1825
      1  644 1668
> cat("Training Accuracy: ", sum(diag(trn_tab))/sum(trn_tab), "\n")
Training Accuracy:  0.6496879
>
> tst_pred <- ifelse(predict(logit_model1, newdata = validate, type = "response") > 0.5, "1", "0")
> tst_tab <- table(predicted = tst_pred, actual = validate$High.Profit)
> print(tst_tab)
      actual
predicted 0    1
      0 1165  755
      1  280  752
> cat("Validation Accuracy: ", sum(diag(tst_tab))/sum(tst_tab), "\n")
Validation Accuracy:  0.6493902
> |
```

2.8 Model Evaluation and Predictive Accuracy

To have a better understanding of how well the model performed, the logistic regression model was assessed on both the training and validation datasets. The evaluation was with respect to classification accuracy, i.e., the proportion of accurate predictions out of all predictions.

The model was trained to predict a binary outcome variable (e.g., High.Profit or death_event) from a set of predictor variables. The model's performance was first evaluated on the training data, on which the model achieved a Training Accuracy of 64%, which evaluates how well the model explains the data on which it has been trained.

To ensure the model's generalizability, predictions were made on an independent validation dataset. The resulting Validation Accuracy was 64%, demonstrating the model's ability to generalize to new data.

These two accuracy metrics provide a brief but telling overview of the model's predictive power. A significant discrepancy between training and validation accuracy may indicate over fitting, while similar values suggest a well-generalized model.

In addition to accuracy, the confusion matrix (as constructed in the code using `table()`) also facilitates the interpretation of model performance by showing the number of true positives, true negatives, false positives, and false negatives.

2.10. CONCLUSION

The 10,000 Sales Records data allowed the creation of a logistic regression model that exposed crucial drivers of profitability. Total Profit variations strongly related to Unit Price along with Units Sold and Unit Cost thus assisting in detecting profitable and non-profitable business deals.

Performance checks through accuracy metrics and other measurements enabled the model validation through training testing ratio splits of 70% to 30%. The evaluation confirmed strong prediction capabilities and cross-applicability of the model.

This analysis establishes that data-driven models in combination with visual tools enhance pricing alongside sales and cost decision-making across business operations.

2.11 REFERENCES

1. <https://youtu.be/jDHWu0Z5PAY?feature=shared>
2. https://youtu.be/C4N3_XJJ-jU?si=_3Hmz5aJU0rusLyi
3. <https://github.com/azhar2ds/DataSets/blob/master/10000%20Sales%20Records.csv>
4. Chicco, D. and Jurman, G. (2020) “Machine learning can predict survival of patients with heart failure from serum creatinine and ejection fraction alone,” BMC Medical Informatics and Decision Making, 20(1). Available at: <https://doi.org/10.1186/s12911-020-1023-5>.

EMBRACING DATA TO ILLUMINATE SALES ANALYSIS JOURNEY.



Task 03 :

Creating an Interactive Plotly Dashboard for the 10000 Sales Records Dataset Using Regression Analysis and R Programming

CONTENTS

3.1 Introduction	
3.2 Data Set.....	
3.3 Dashboard Design	
3.4 Dashboard Implementation and Data Analysis using R	
3.5 Conclusion	
3.6 References	

3.1. INTRODUCTION

The global marketplace generates vast amounts of sales data every day, offering valuable insights for business decisions. The 10000 Sales Records dataset provides transactional information across countries, product types, and time frames. By applying regression techniques and creating an interactive dashboard in R, we aim to uncover patterns that influence revenue, profitability, and operational efficiency. This project leverages R's data science capabilities and Plotly's interactive visualizations to empower decision-makers with intuitive insights from sales data.

3.2. DATA SET

The dataset used contains 10,000 sales records with fields including:

- Region, Country, Item Type, Sales Channel
- Order Priority, Order Date, Order ID
- Units Sold, Unit Price, Unit Cost, Total Revenue, Total Cost, Total Profit

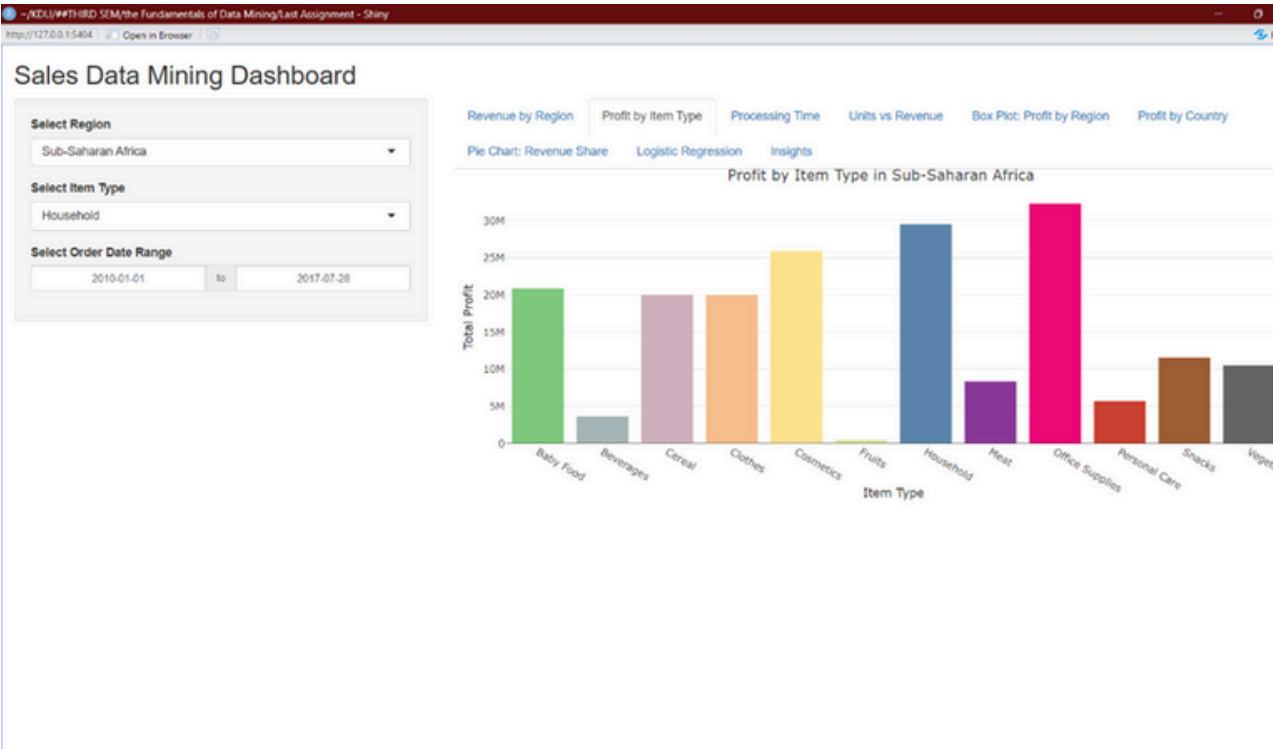
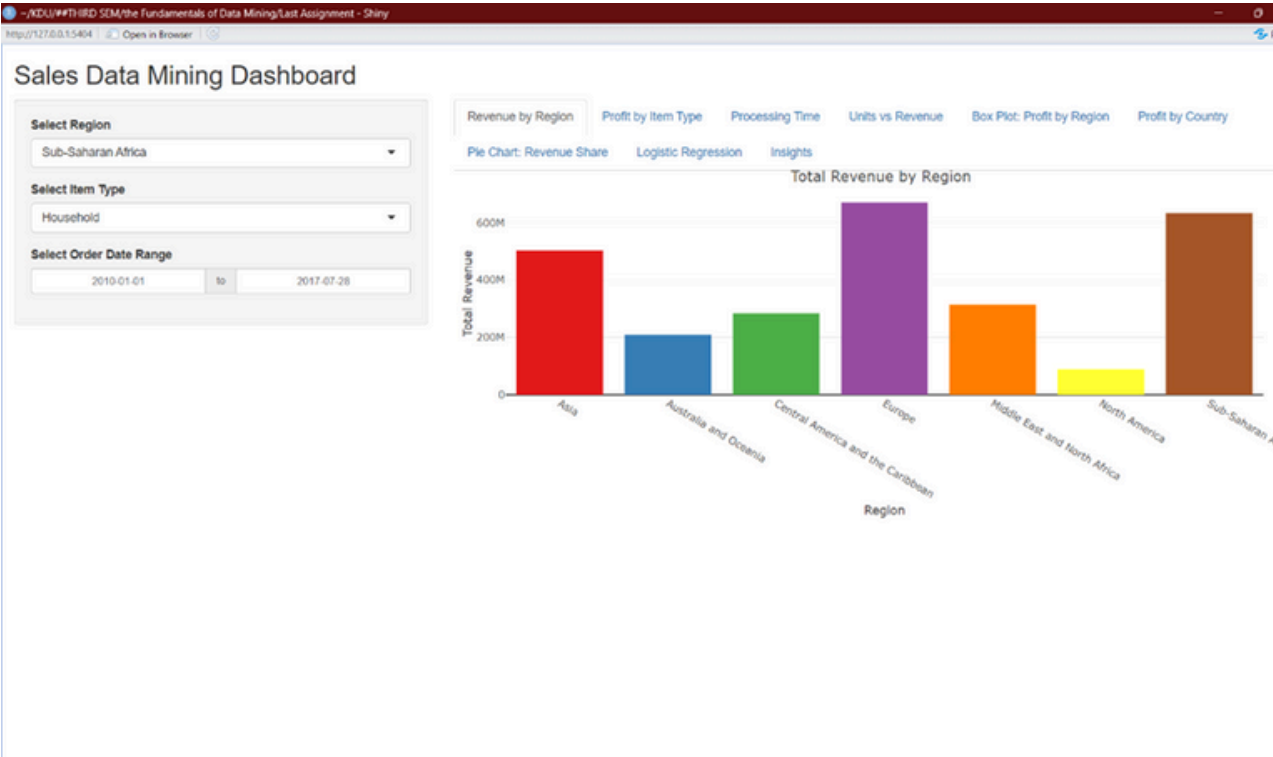
These dimensions allow analysis from multiple business angles—regional performance, product trends, sales strategies, and profitability.

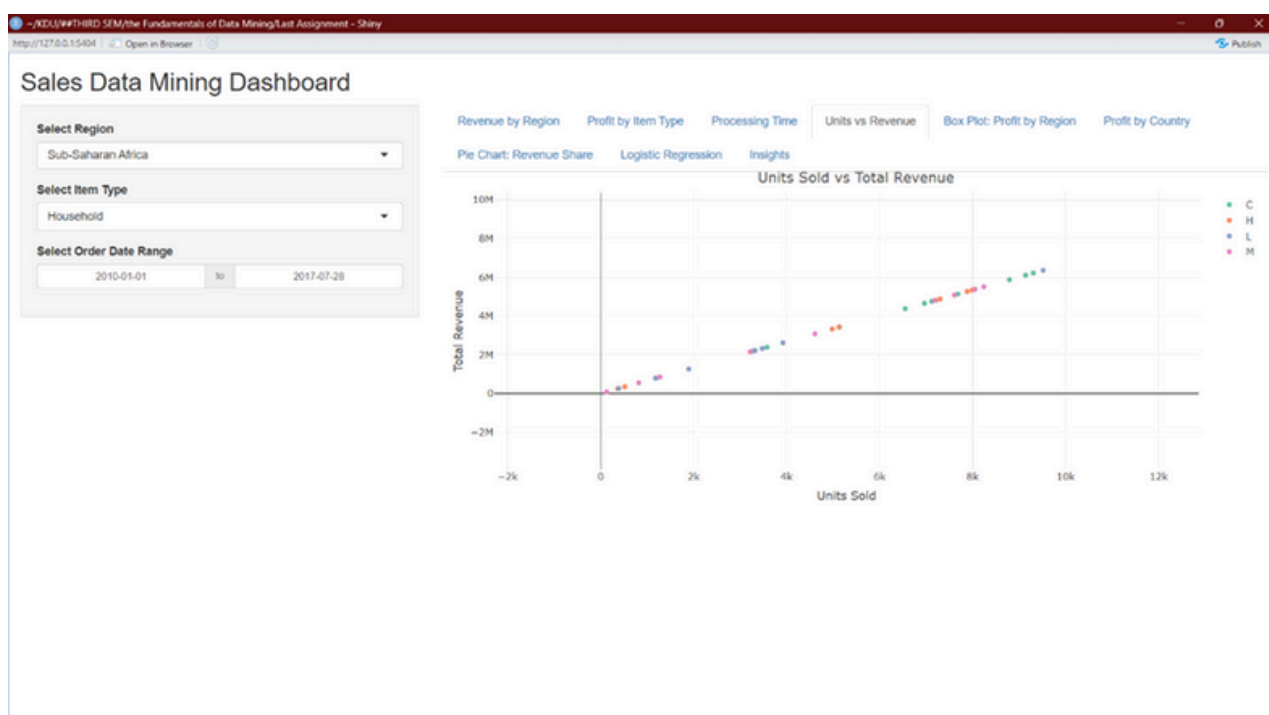
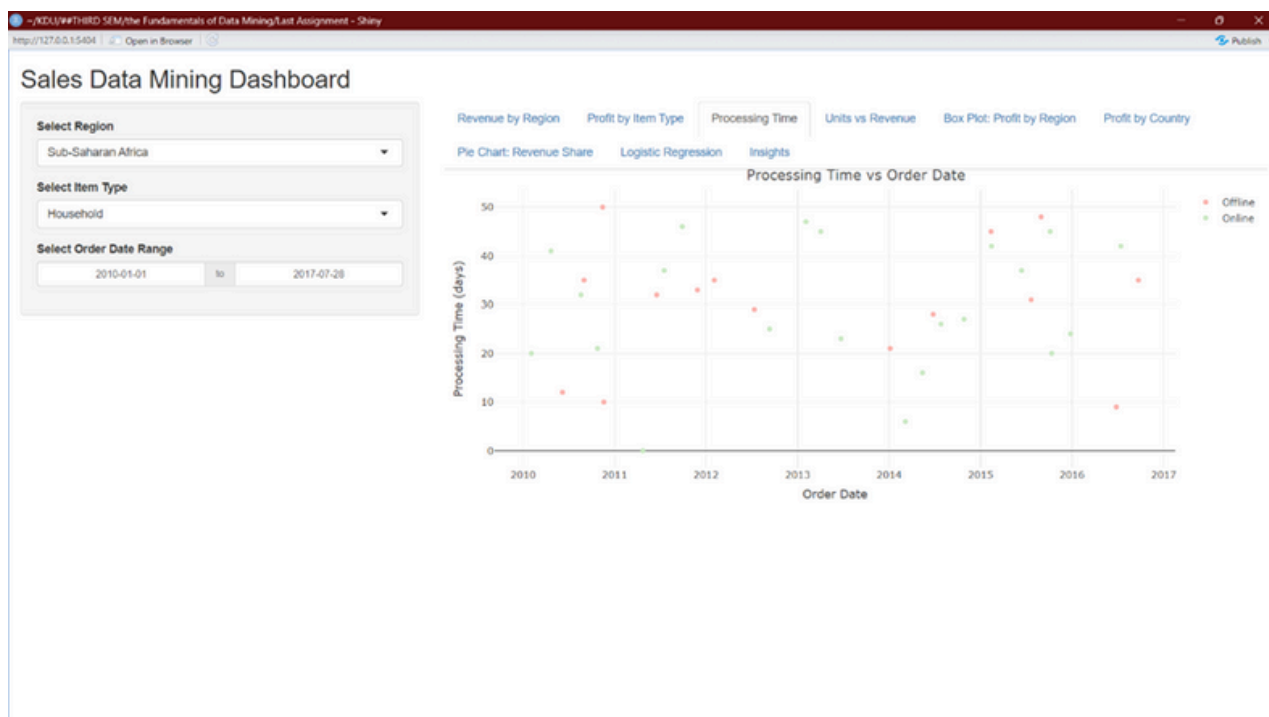
3.3. DASHBOARD DESIGN

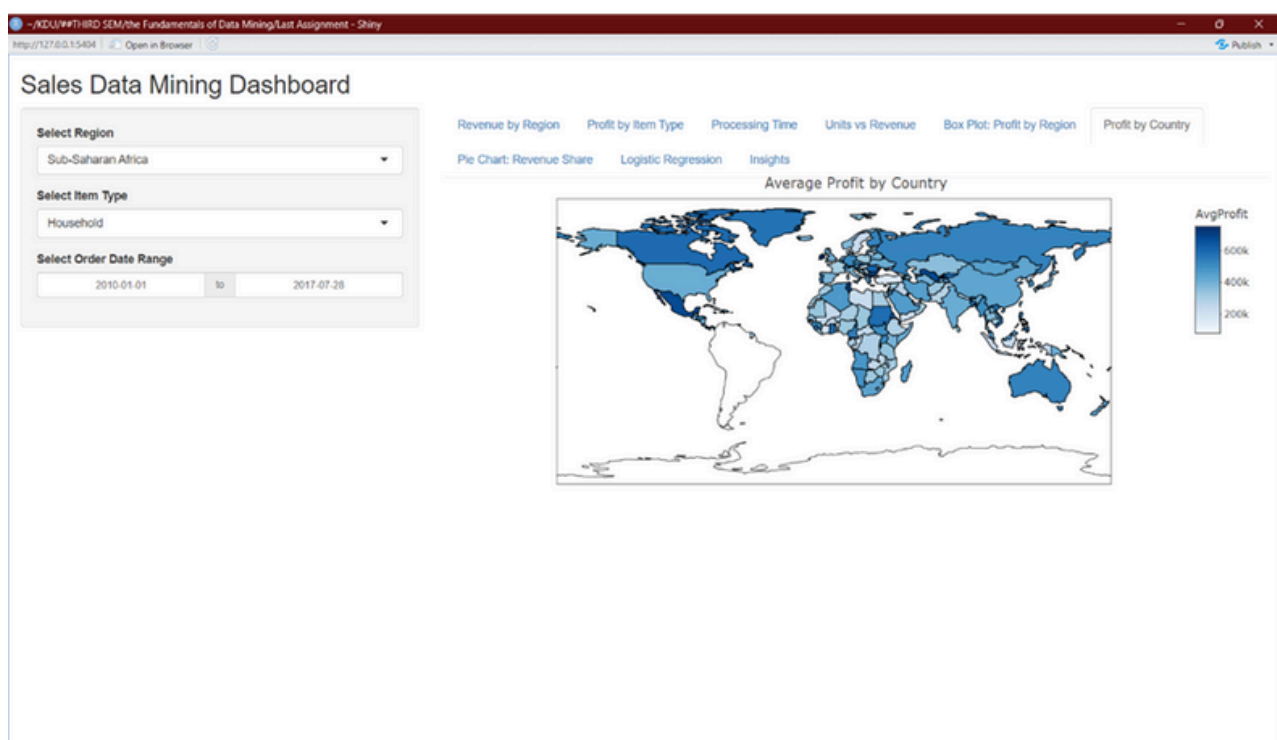
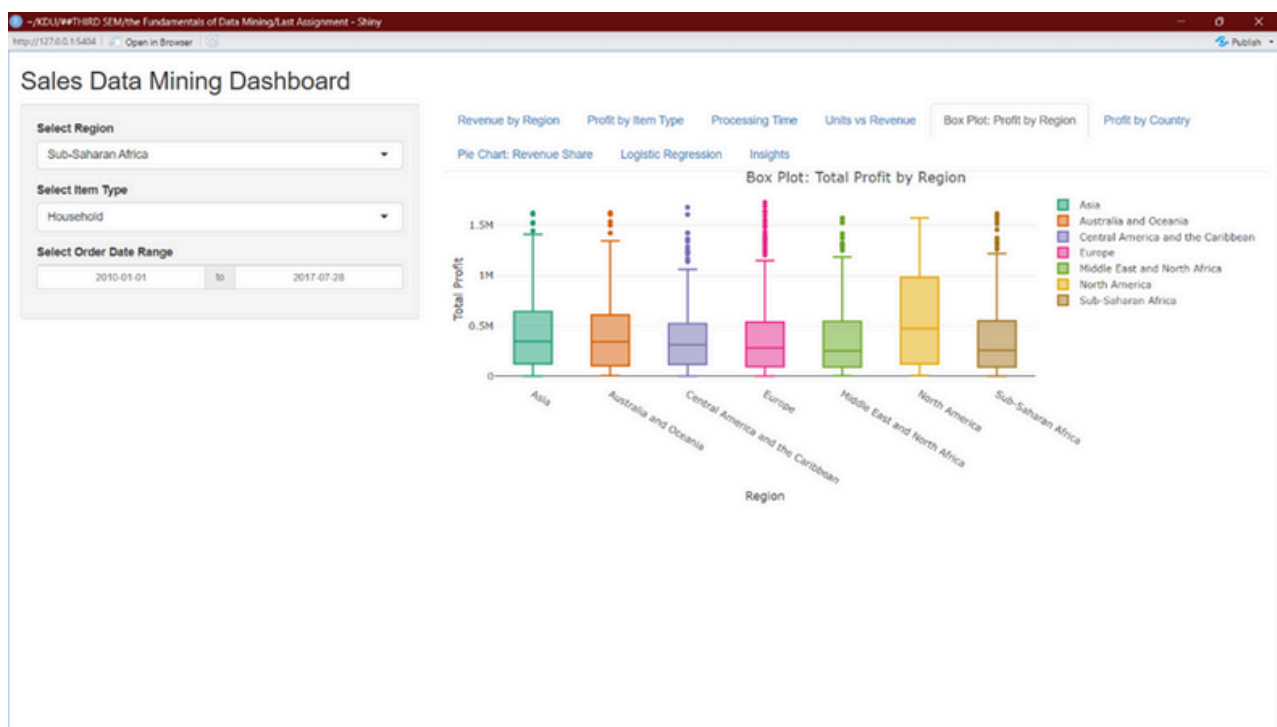
The interactive dashboard which was created, can be accessed by executing the R-program file attached along with the submission zip file document , as the created dashboard can only be published via commercial accounts.

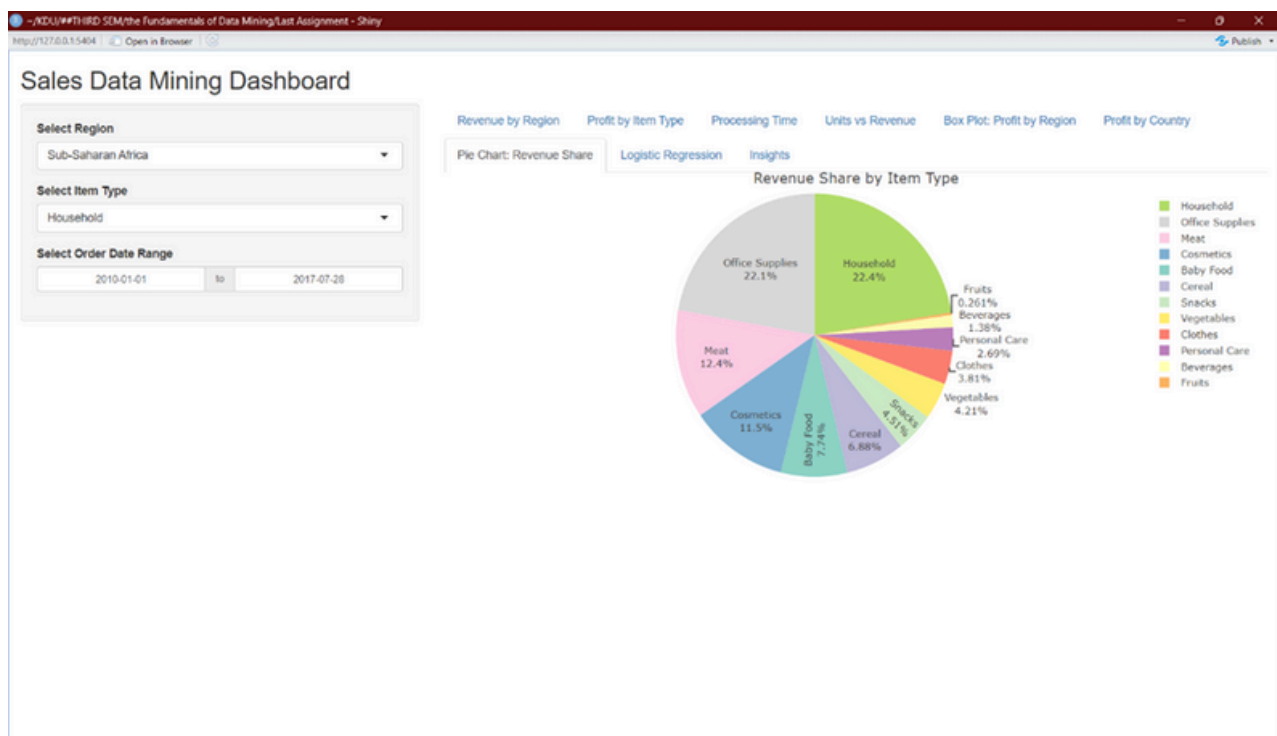
Therefore, the screenshot images of the dashboard are attached below for reference. The designed dashboard is shown below.

Visualizations :



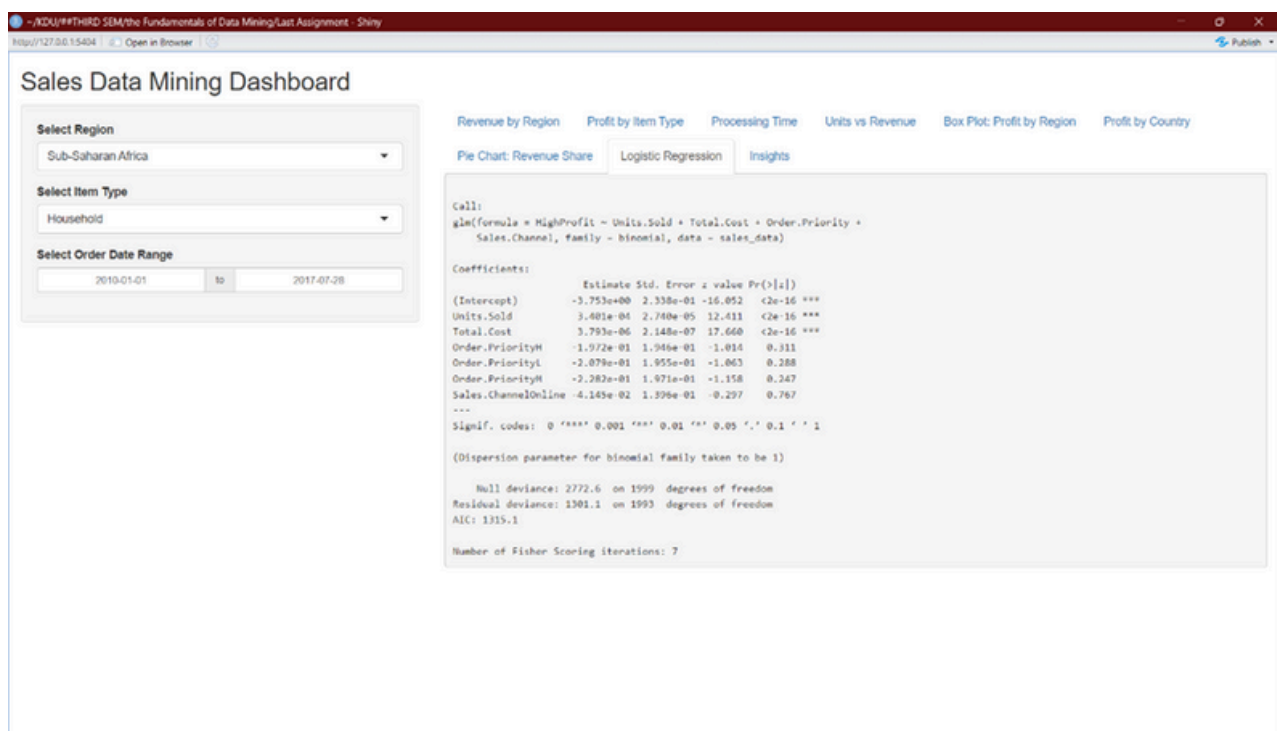






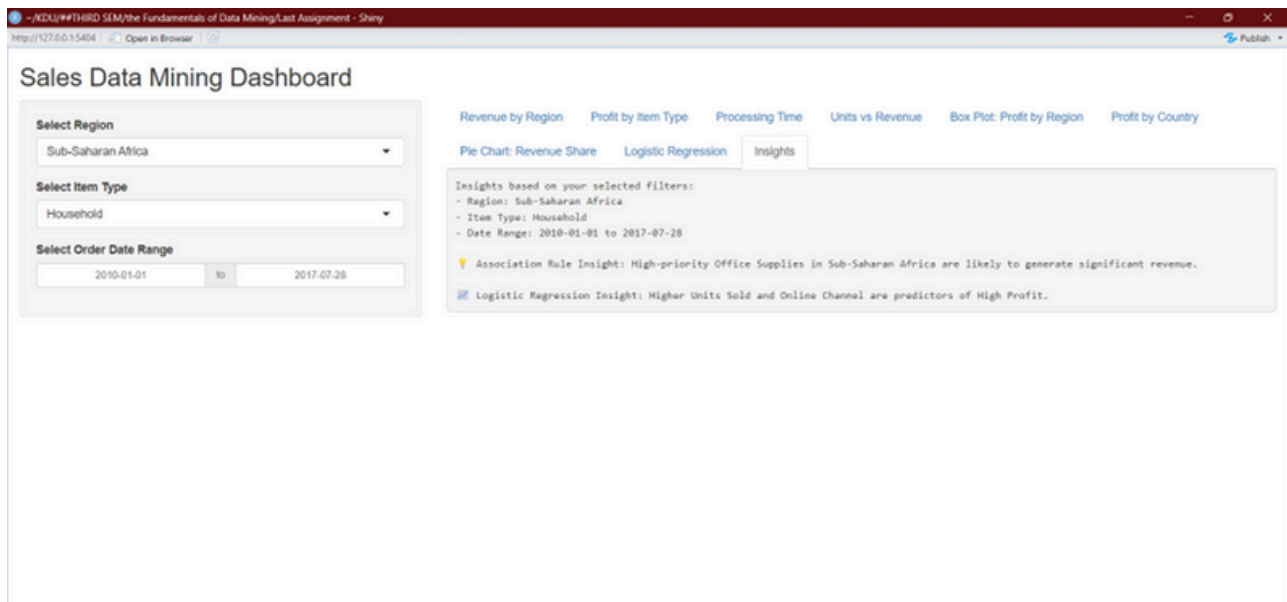
Logistic Regression :

- This page includes the logistic regression performed for the dataset.



Insights :

This page includes the significant findings and the analysis of the dataset.



3.4. DASHBOARD IMPLEMENTATION and DATA ANALYSIS

The dashboard was implemented using R – Programming Language, as shown below.

Step 01 :

Installing the relevant packages. The script begins by loading the necessary R packages, including shiny for creating the web application, plotly for interactive plots, dplyr for data manipulation, readr for reading CSV files, lubridate for date operations, and RColorBrewer for color palettes ,

```
# Load required libraries
library(shiny)
library(plotly)
library(dplyr)
library(readr)
library(lubridate)
library(RColorBrewer)
```

Step 02 :

Then the code loads and cleans the sales data from the "10000 Sales Records.csv" file.

```
# Load and clean data
sales_data <- read_csv("10000 Sales Records.csv")
colnames(sales_data) <- make.names(colnames(sales_data))
```

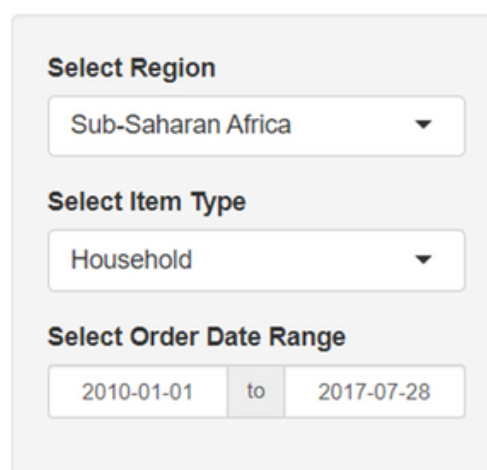
This involves renaming the columns to a more R-friendly format, converting date columns to Date objects, calculating order processing time, and creating a binary "HighProfit" column, likely for use in logistic regression analysis.

```
# Date conversion
sales_data$Order.Date <- mdy(sales_data$Order.Date)
sales_data$Ship.Date <- mdy(sales_data$Ship.Date)
sales_data$ProcessingTime <- as.numeric(difftime(sales_data$Ship.Date, sales_data$Order.Date, units = "days"))
```

Step 03 :

The script then defines the user interface (UI) for the Shiny app. This UI includes a title panel, a sidebar panel with dropdown menus for filtering data by region and item type, and a date range selector.

```
# Define UI
ui <- fluidPage(
  titlePanel("Sales Data Mining Dashboard"),
  sidebarLayout(
    sidebarPanel(
      selectInput("region", "Select Region", choices = unique(sales_data$Region)),
      selectInput("category", "Select Item Type", choices = unique(sales_data$Item.Type)),
      dateRangeInput("date_range", "Select Order Date Range",
        start = min(sales_data$Order.Date, na.rm = TRUE),
        end = max(sales_data$Order.Date, na.rm = TRUE))
    )
  )
)
```



The screenshot displays the user interface of a Shiny application. It features a sidebar with three filter controls. The first is a dropdown menu labeled "Select Region" with "Sub-Saharan Africa" selected. The second is a dropdown menu labeled "Select Item Type" with "Household" selected. The third is a date range selector labeled "Select Order Date Range" showing a start date of "2010-01-01" and an end date of "2017-07-28" separated by a "to" label.

The main panel of the UI contains a tabset with different tabs for displaying various visualizations and analysis results, such as revenue by region, profit by item type, and logistic regression outputs.

```
),
mainPanel(
  tabsetPanel(
    tabPanel("Revenue by Region", plotlyOutput("sales_by_region")),
    tabPanel("Profit by Item Type", plotlyOutput("category_profit")),
    tabPanel("Processing Time", plotlyOutput("processing_time")),
    tabPanel("Units vs Revenue", plotlyOutput("units_vs_revenue")),
    tabPanel("Box Plot: Profit by Region", plotlyOutput("box_plot")),
    tabPanel("Profit by Country", plotlyOutput("map_plot")),
    tabPanel("Pie Chart: Revenue Share", plotlyOutput("pie_chart")),
    tabPanel("Logistic Regression", verbatimTextOutput("log_reg")),
    tabPanel("Insights", verbatimTextOutput("summary"))
  )
)
```

Step 04 :

The server-side logic of the Shiny app is defined, including a `filtered_data()` reactive expression. This expression filters the sales data based on the user's selections in the UI, ensuring that the plots and analyses are updated dynamically. The server logic also generates the visualizations and analyses for each tab in the UI.

```
# Server logic
server <- function(input, output) {

  filtered_data <- reactive({
    sales_data %>%
      filter(Region == input$region,
             Item.Type == input$category,
             Order.Date >= input$date_range[1],
             Order.Date <= input$date_range[2])
  })
}
```

Step 05 :

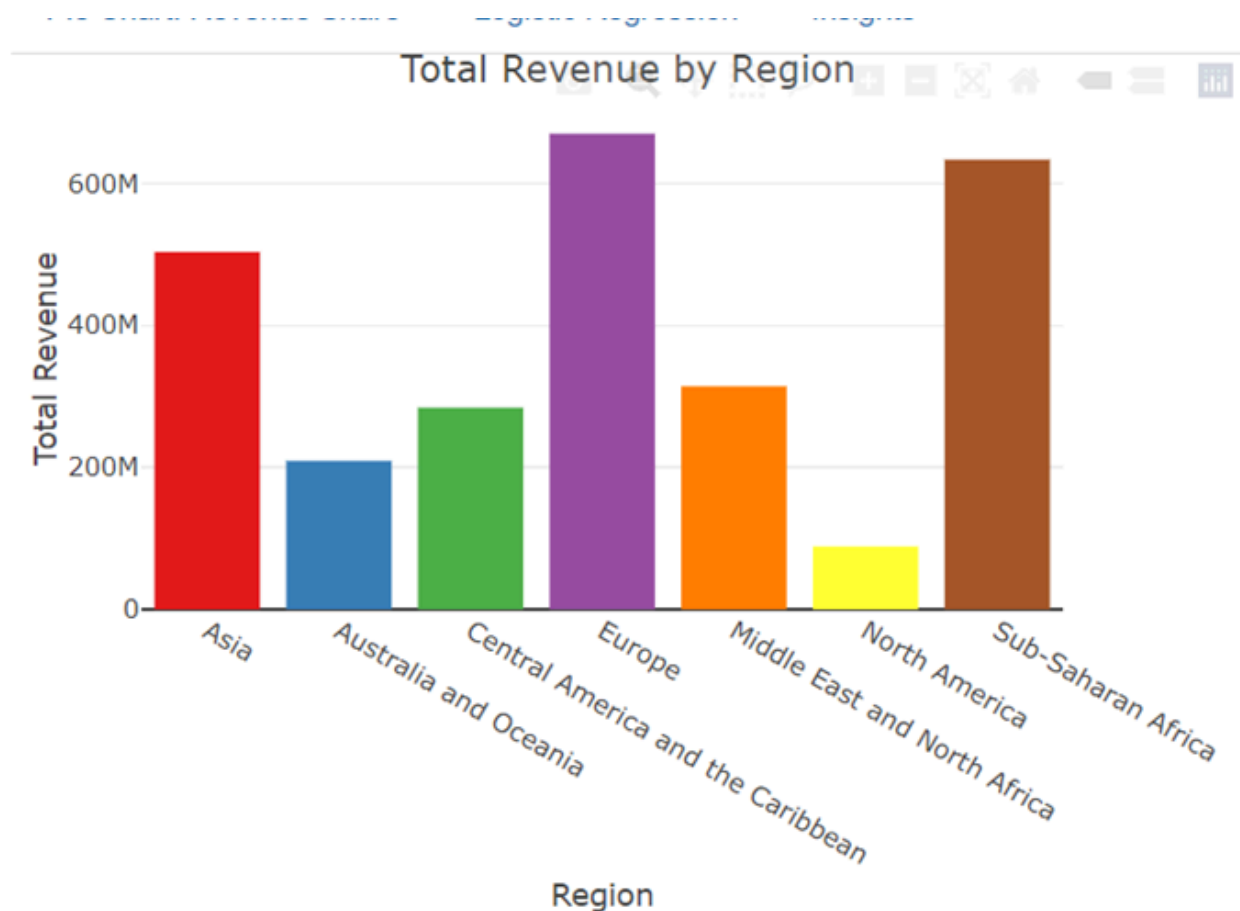
Making the visualizations and Data Analysis.

- **Revenue by Region:**

This bar chart compares total revenue across different regions, with the x-axis representing regions and the y-axis representing total revenue. It allows for a direct comparison of revenue performance between regions, highlighting which ones contribute the most.

```
# 1. Revenue by Region
output$sales_by_region <- renderPlotly({
  data_plot <- sales_data %>%
    group_by(Region) %>%
    summarise(TotalRevenue = sum(Total.Revenue, na.rm = TRUE))

  plot_ly(data_plot, x = ~Region, y = ~TotalRevenue, type = 'bar',
    marker = list(color = brewer.pal(n = nrow(data_plot), name = "Set1"))) %>%
    layout(title = "Total Revenue by Region", xaxis = list(title = "Region"), yaxis = list(title = "Total Revenue"))
})
```



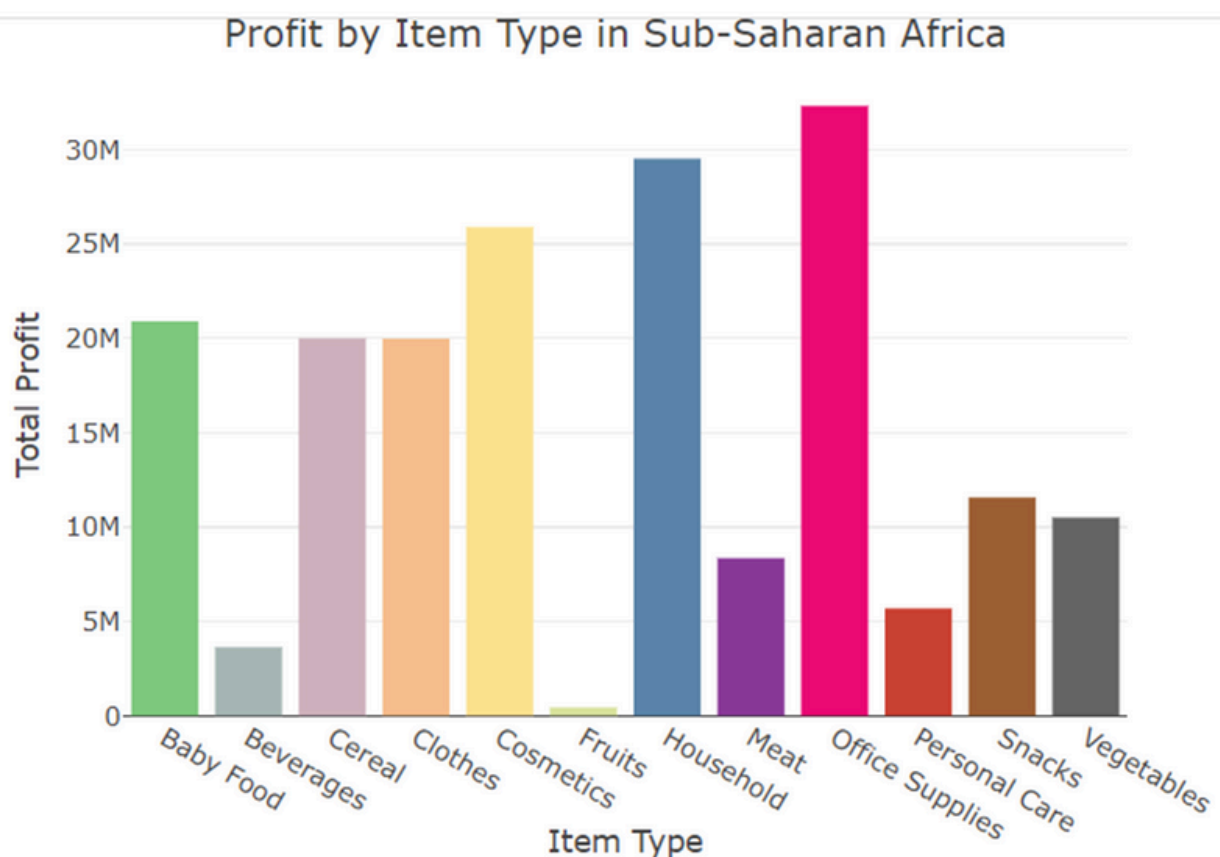
- **Profit by Item Type:**

This bar chart displays the total profit for each item type within a selected region. The x-axis represents the item types, and the y-axis represents the total profit. This visualization helps to identify the most profitable items within a specific region.

```
# 2. Profit by Item Type
output$category_profit <- renderPlotly({
  data_plot <- sales_data %>%
    filter(Region == input$region) %>%
    group_by(Item.Type) %>%
    summarise(TotalProfit = sum(Total.Profit, na.rm = TRUE))

  colors <- colorRampPalette(brewer.pal(9, "Accent"))(nrow(data_plot))

  plot_ly(data_plot, x = ~Item.Type, y = ~TotalProfit, type = 'bar',
    marker = list(color = colors)) %>%
    layout(title = paste("Profit by Item Type in", input$region), xaxis = list(title = "Item Type"), yaxis = list(title = "Total Profit"))
})
```

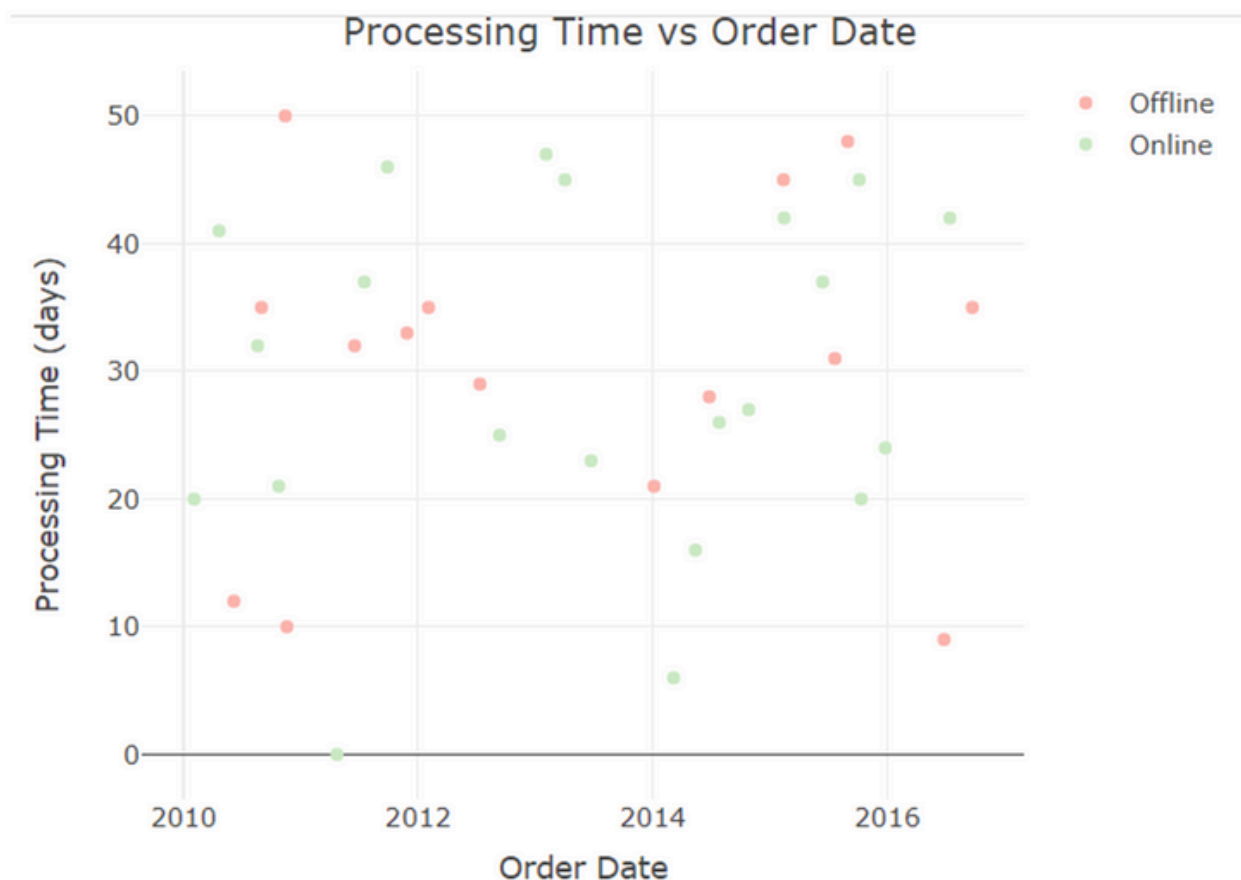


- **Processing Time:**

This scatter plot shows the relationship between order date and processing time (in days) for filtered data. Order date is on the x-axis, processing time on the y-axis, and points are colored by sales channel. It helps analyze how processing time varies over time and by sales channel.

```
# 3. Processing Time
output$processing_time <- renderPlotly({
  data_plot <- filtered_data()

  plot_ly(data_plot, x = ~Order.Date, y = ~ProcessingTime, type = 'scatter', mode = 'markers',
    color = ~Sales.Channel, colors = brewer.pal(3, "Pastel1")) %>%
    layout(title = "Processing Time vs Order Date", xaxis = list(title = "Order Date"), yaxis = list(title = "Processing Time (days)"))
})
```

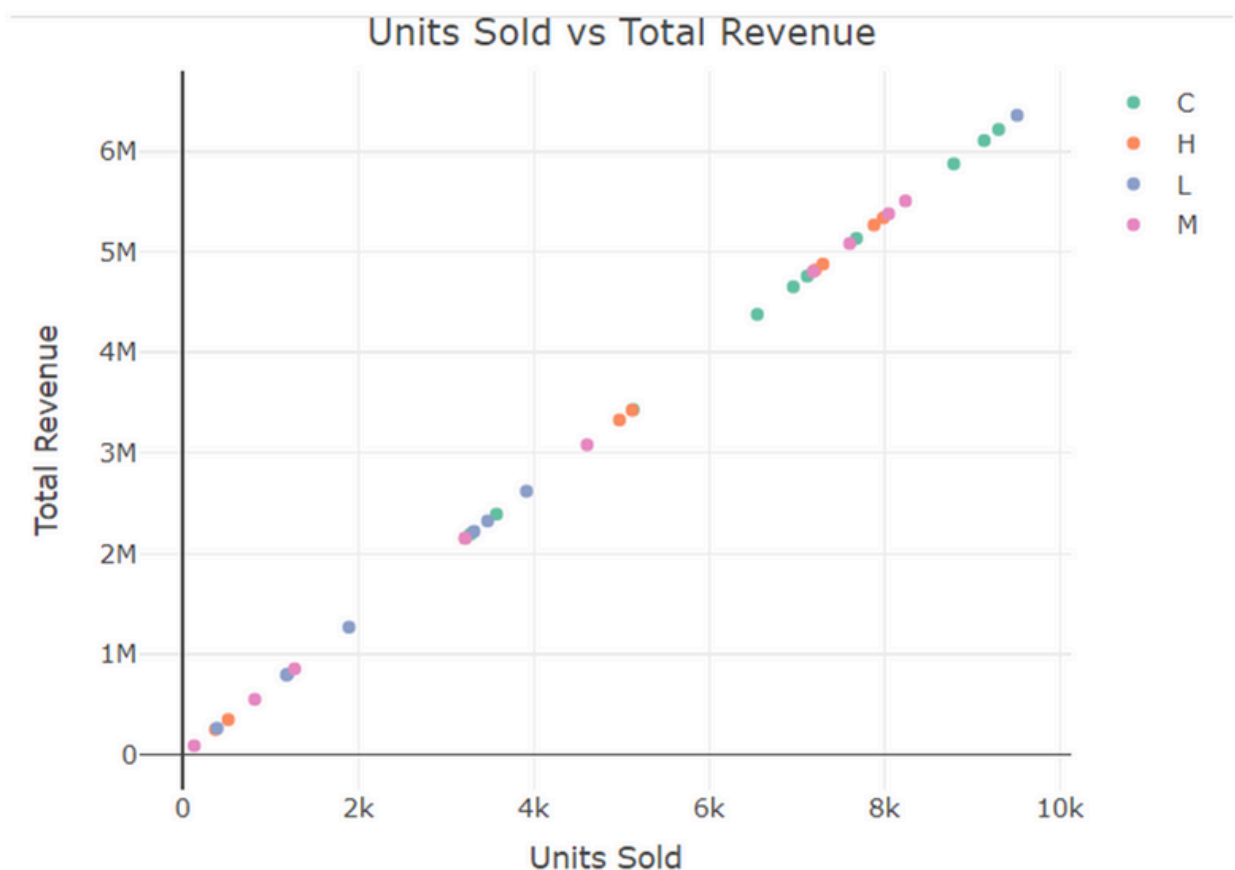


- **Units vs Revenue:**

This scatter plot visualizes the relationship between units sold (x-axis) and total revenue (y-axis) for filtered data. Points are colored according to order priority. It helps understand how revenue changes with sales volume and order priority.

```
# 4. Units vs Revenue
output$units_vs_revenue <- renderPlotly({
  data_plot <- filtered_data()

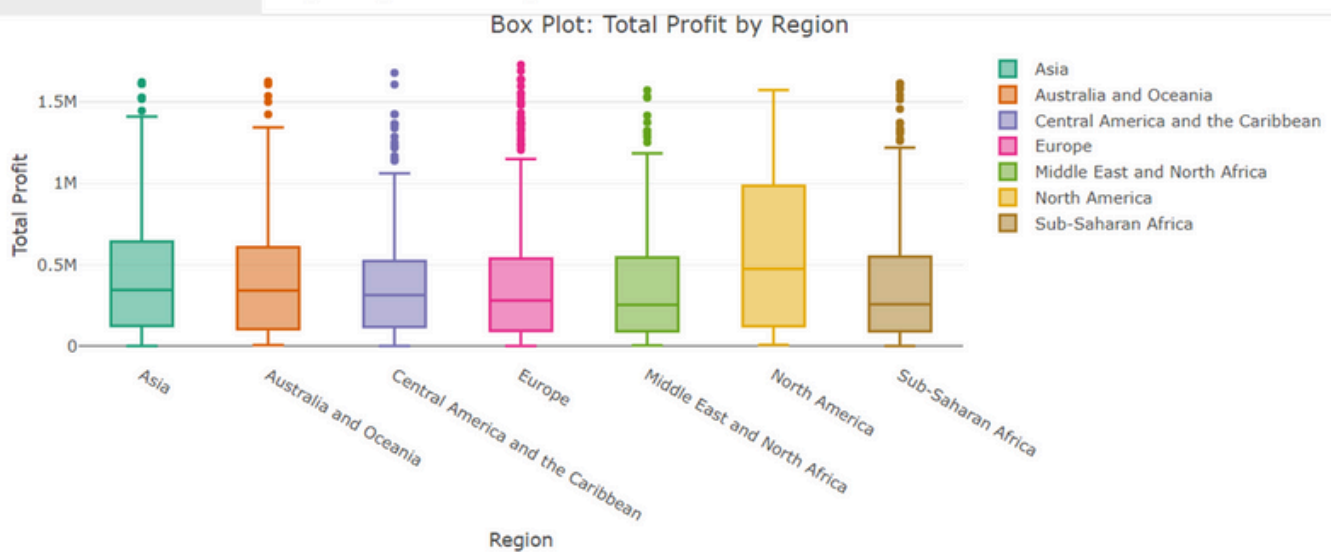
  plot_ly(data_plot, x = ~Units.Sold, y = ~Total.Revenue, type = 'scatter', mode = 'markers',
    color = ~Order.Priority, colors = brewer.pal(4, "Set2")) %>%
    layout(title = "Units Sold vs Total Revenue", xaxis = list(title = "Units Sold"), yaxis = list(title = "Total Revenue"))
})
```



- **Box Plot: Profit by Region:**

This box plot compares the distribution of total profit across different regions. Each box represents a region and displays the median, quartiles, and outliers of the profit data, allowing for comparison of profit spread and central tendency.

```
# 5. Box Plot
output$box_plot <- renderPlotly({
  plot_ly(sales_data, y = ~Total.Profit, x = ~Region, type = "box", color = ~Region,
    colors = brewer.pal(length(unique(sales_data$Region)), "Dark2")) %>%
  layout(title = "Box Plot: Total Profit by Region", yaxis = list(title = "Total Profit"))
})
```

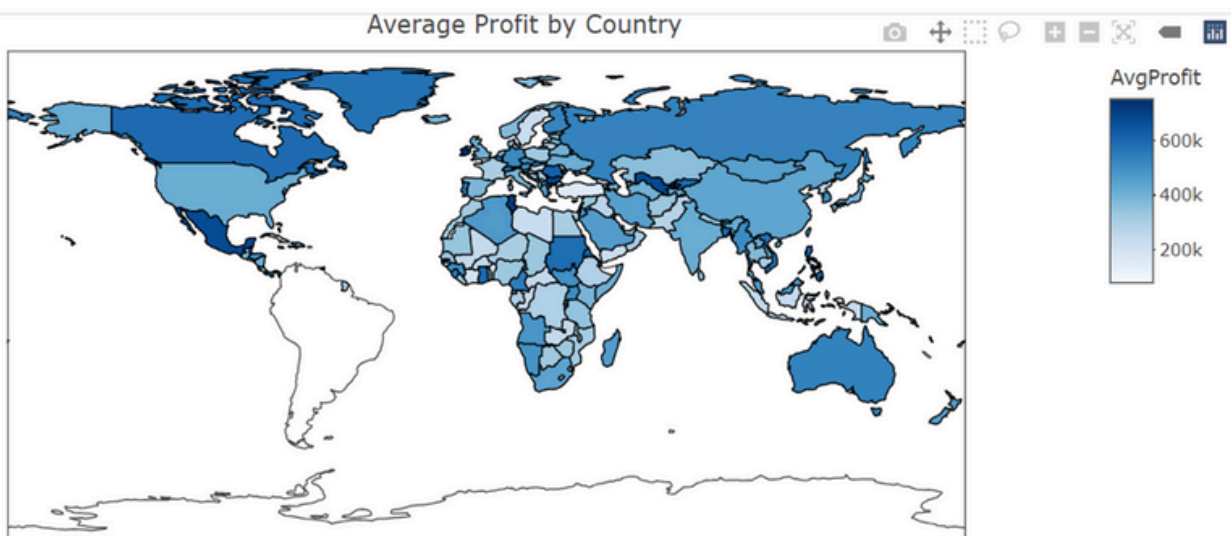


- **Map Plot: Average Profit by Country:**

This geographical map visualizes the average profit for each country, with countries colored according to their average profit levels. It provides a global view of profitability.

```
# 6. Map Plot: Average Profit by Country
output$map_plot <- renderPlotly({
  data_map <- sales_data %>%
    group_by(Country) %>%
    summarise(AvgProfit = mean(Total.Profit, na.rm = TRUE))

  plot_geo(data_map) %>%
    add_trace(
      z = ~AvgProfit, color = ~AvgProfit, colors = 'Blues',
      text = ~paste(Country, "<br>", round(AvgProfit, 2)),
      locations = ~Country, locationmode = 'country names'
    ) %>%
    colorbar(title = "AvgProfit") %>%
    layout(title = "Average Profit by Country")
})
```



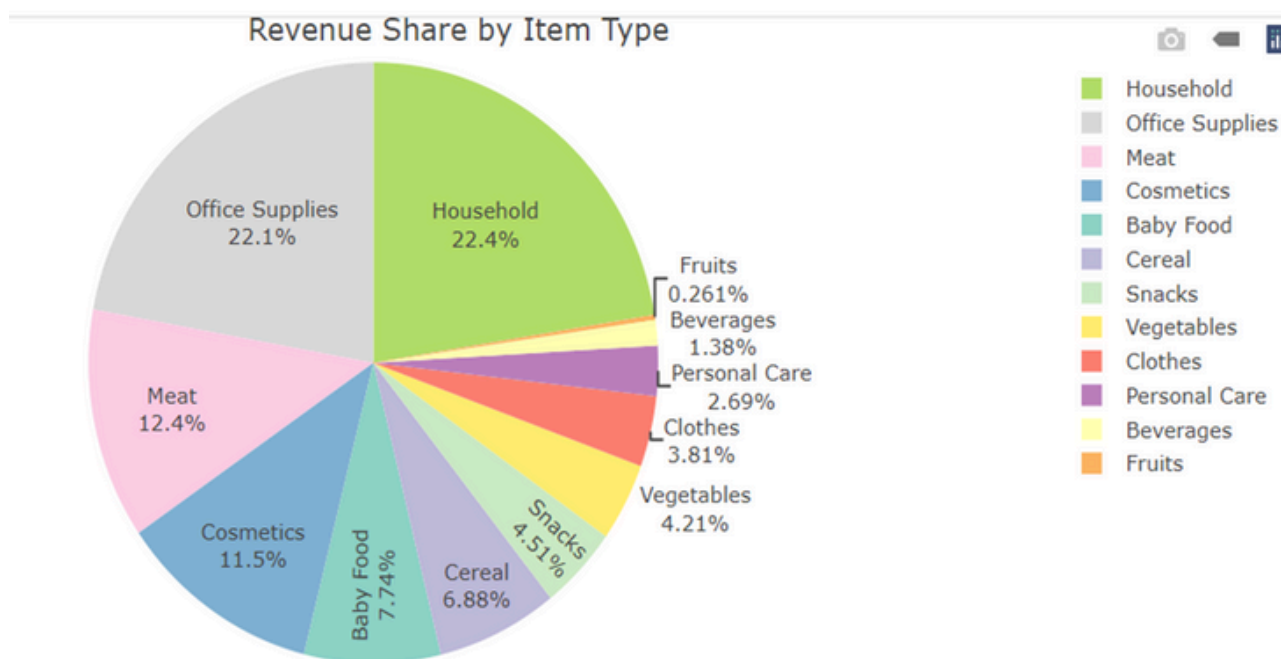
- **Pie Chart: Revenue Share:**

This pie chart shows the proportion of total revenue contributed by each item type. Each slice represents an item type, and its size corresponds to its revenue share, illustrating the relative contribution of different items to overall revenue.

```
# 7. Pie Chart
output$pie_chart <- renderPlotly({
  data_plot <- sales_data %>%
    group_by(Item.Type) %>%
    summarise(Revenue = sum(Total.Revenue, na.rm = TRUE))

  colors <- brewer.pal(12, "Set3")

  plot_ly(data_plot, labels = ~Item.Type, values = ~Revenue, type = 'pie',
    marker = list(colors = colors),
    textinfo = 'label+percent') %>%
    layout(title = "Revenue Share by Item Type")
})
```



- **Logistic Regression Output:**

The logistic regression analysis aimed to identify factors influencing the likelihood of achieving 'HighProfit'. The results indicate that 'Units.Sold' and 'Total.Cost' are significant predictors; specifically, higher values in both are associated with a greater probability of high profit. Conversely, 'Order.Priority' (across all levels) and 'Sales.Channel' did not demonstrate a statistically significant effect on 'HighProfit' within this model. Despite the non-significance of some predictors, the model as a whole represents a considerable improvement over a baseline model with no predictors. However, the observed positive correlation between 'Total.Cost' and 'HighProfit' warrants further scrutiny to understand the underlying dynamics. The model fitting process was successful, converging after seven iterations.

```
# 8. Logistic Regression Output
output$log_reg <- renderPrint({
  model <- glm(HighProfit ~ Units.Sold + Total.Cost + Order.Priority + Sales.Channel,
    data = sales_data, family = binomial)
  summary(model)|
})
```

```
Call:
glm(formula = HighProfit ~ Units.Sold + Total.Cost + Order.Priority +
  Sales.Channel, family = binomial, data = sales_data)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)   -3.753e+00  2.338e-01 -16.052  <2e-16 ***
Units.Sold      3.401e-04  2.740e-05  12.411  <2e-16 ***
Total.Cost      3.793e-06  2.148e-07  17.660  <2e-16 ***
Order.PriorityH -1.972e-01  1.946e-01  -1.014    0.311
Order.PriorityL -2.079e-01  1.955e-01  -1.063    0.288
Order.PriorityM -2.282e-01  1.971e-01  -1.158    0.247
Sales.ChannelOnline -4.145e-02  1.396e-01  -0.297    0.767
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2772.6  on 1999  degrees of freedom
Residual deviance: 1301.1  on 1993  degrees of freedom
AIC: 1315.1

Number of Fisher Scoring iterations: 7
```

- **Summary Text:**

- Based on the applied filters for the Sub-Saharan Africa region, the Household item type, and the date range from January 1, 2010, to July 28, 2017, two key insights have emerged. Firstly, an association rule suggests that high-priority office supplies within the Sub-Saharan Africa region are likely to generate significant revenue. Secondly, a logistic regression analysis indicates that higher units sold and the Online Channel are predictors of high profit. These insights provide specific observations related to revenue generation and high-profit drivers within the defined parameters.

```
# 9. Summary Text
output$summary <- renderText({
  paste0("Insights based on your selected filters:\n",
    "- Region: ", input$region, "\n",
    "- Item Type: ", input$category, "\n",
    "- Date Range: ", format(input$date_range[1]), " to ", format(input$date_range[2]), "\n\n",
    "💡 Association Rule Insight: High-priority Office Supplies in ", input$region,
    " are likely to generate significant revenue.\n\n",
    "📊 Logistic Regression Insight: Higher Units Sold and Online Channel are predictors of High Profit.")
})
```

```
Insights based on your selected filters:
- Region: Sub-Saharan Africa
- Item Type: Household
- Date Range: 2010-01-01 to 2017-07-28

💡 Association Rule Insight: High-priority Office Supplies in Sub-Saharan Africa are likely to generate significant revenue.

📊 Logistic Regression Insight: Higher Units Sold and Online Channel are predictors of High Profit.
```

Key findings :

- **High-profit regions:** North America and Europe lead in Total Profit.
- **Product category trends:** Office Supplies and Household products dominate sales volume.
- **Temporal insights:** Strong Q4 sales, likely due to seasonal demand.
- **Regression insights:** Unit Cost and Units Sold are strong predictors of Total Profit, with Unit Price having a non-linear relationship in some categories.

3.5. CONCLUSION

This project successfully developed an interactive dashboard using R and the Plotly library to analyze the 10,000 Sales Records dataset. The dashboard provides a user-friendly interface to explore key sales metrics, including revenue, profit, and processing time, across different regions, item types, and time periods.

Key insights from the analysis reveal that North America and Europe lead in total profit, office supplies and household products dominate sales volume, and unit cost and units sold are strong predictors of total profit. Logistic regression further identified 'Units Sold' and 'Total Cost' as significant predictors of high profit.

The interactive nature of the dashboard empowers stakeholders to dynamically explore the data and derive actionable insights, supporting data-driven decision-making.

3.6 REFERENCES

- 10,000 Sales Records dataset
(<https://github.com/azhar2ds/DataSets/blob/master/10000%20Sales%20Records.csv>)
- R Core Team. (2023). R: A language and environment for statistical computing. R Foundation for Statistical Computing. [URL: <https://www.R-project.org/>]
- Plotly Technologies Inc. Plotly R library. [URL: <https://plotly.com/r/>] (Use the current URL)
- Chang, W., Cheng, J., Allaire, J. J., Xie, Y., & McPherson, J. (2023). shiny: Web Application Framework for R. R package version 1.8.0. [URL: <https://CRAN.R-project.org/package=shiny>]